
Optimal Real-Data Allocation for Synthetic-Data-Augmented Inference

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Abstract

1 Large language models have made synthetic data inexpensive, but they can still
2 be biased. When real data are scarce, researchers must decide how much of
3 the real sample to use for direct estimation and how much to spend calibrating
4 the generator. We study this allocation problem and derive the marginal-value
5 condition for the optimal calibration size: $(v_n + cx^{-2\beta})^2 = 2\beta acx^{-(2\beta+1)}$, where
6 x is the calibration size, v_n is synthetic sampling variance, a is the real-estimator
7 variance constant, and $cx^{-2\beta}$ is squared synthetic bias. This condition has no
8 universal fixed-share solution and yields five allocation regimes. In the main regime,
9 $m_n \asymp n$ and $\beta > 1/2$, the optimal calibration size grows only as $n^{2/(2\beta+1)}$, so
10 the calibration share vanishes; rules that keep a positive fixed fraction of real
11 observations in calibration over-calibrate asymptotically. We also propose an
12 adaptive grid estimator with an oracle inequality and a safety check, $xB_n(x) < a$,
13 that falls back to all-real estimation when synthetic data cannot help. Across
14 approximately 10^7 Monte Carlo replications, the corrected oracle reduces MSE by
15 about 47% relative to all-real estimation in the headline cell, compared with about
16 22% for a fixed-share baseline. For inference, naive Wald intervals around the
17 MSE-optimal estimator can undercover, while validation debiasing substantially
18 restores coverage. These results provide guidance for when synthetic data are
19 useful, how much real data should be used to make them reliable, and how to
20 conduct valid inference with synthetic-data-augmented estimators.

21 1 Introduction

22 Large language models have made synthetic data cheap. A single generator can now produce millions
23 of labeled examples, simulated outcomes, or auxiliary observations on demand [Kaplan et al., 2020,
24 Hoffmann et al., 2022]. This is valuable when real data are expensive, sensitive, slow to collect,
25 or constrained by institutions. But abundance does not remove the empirical bottleneck: synthetic
26 observations come from a model, and that model may preserve bias, misspecified structure, or
27 distortions inherited from training and alignment. The central question is therefore not whether
28 synthetic data can be generated, but how much scarce real data should be spent making them reliable.

29 We study this question as a real-data allocation problem. A researcher wants to estimate a scalar
30 parameter θ^* of a real population, such as a treatment effect, a mean, or a regression coefficient,
31 and has a budget of n real observations. Each observation can be used as direct evidence about
32 θ^* or as calibration data that makes future synthetic draws less biased. Calibration improves the
33 artificial data-generating process, but it also reduces the real sample left for direct estimation. This is
34 a Neyman-style allocation problem [Neyman, 1934], except the split is across uses of data rather than
35 across strata.

36 A tempting shortcut is to choose the calibration share as a fixed function of the bias-decay exponent.
 37 That shortcut misses a second force in the problem. Under a power-law squared-bias curve $b_2(x) \asymp$
 38 $cx^{-2\beta}$, where x is the calibration size and β is the bias-decay exponent, the synthetic estimator also
 39 has sampling variance $v_n = \sigma_S^2/m_n$. The relevant synthetic error is therefore $B_n(x) = v_n + cx^{-2\beta}$,
 40 not the bias term alone.

41 Keeping this term changes the allocation logic. The researcher must ask whether one more calibration
 42 observation reduces synthetic error enough to justify the lost real-estimation precision. This marginal-
 43 value comparison yields the first-order condition

$$(v_n + cx^{-2\beta})^2 = 2\beta acx^{-(2\beta+1)}.$$

44 When the synthetic budget grows with the real budget, $m_n \asymp n^\rho$, and generator bias declines faster
 45 than the parametric rate, $\beta > 1/2$, this condition often calls for sublinear calibration. In the leading
 46 case $m_n \asymp n$, the optimal calibration share goes to zero even though the number of calibration
 47 observations diverges. Positive fixed-share rules therefore over-calibrate in this regime.

48 Existing literatures stop one step before the allocation problem created by synthetic data. Neyman
 49 allocation optimizes sampling across strata [Neyman, 1934, Cochran, 1977]; here the split is across
 50 *uses* of the same scarce real observations. Shrinkage and model-combination methods choose weights
 51 after estimators exist [Stein, 1956, James and Stein, 1961, Hansen, 2007]; they do not decide how
 52 much real data should be spent making a biased synthetic estimator usable. Double machine learning
 53 and prediction-powered inference correct bias under fixed splitting rules [Chernozhukov et al., 2018,
 54 Angelopoulos et al., 2023, Zrnic and Candès, 2024]; they do not supply the marginal-value condition
 55 for calibration itself. Transfer and auxiliary-data learning treat the outside source as given [Pan and
 56 Yang, 2010, Bastani, 2021, Li et al., 2022], while neural scaling laws describe how models improve
 57 with data [Kaplan et al., 2020, Hoffmann et al., 2022]. Our contribution is to turn these pieces into a
 58 real-data allocation theory: a corrected first-order condition, its phase diagram, and inference rules
 59 built on standard semiparametric machinery [Bickel et al., 1993, van der Vaart, 1998].

60 Our contributions are as follows:

- 61 • **Corrected allocation condition.** For any interior optimum x^* of the profiled mean-squared error,
 62 the calibration size satisfies $(v_n + cx^{-2\beta})^2 = 2\beta acx^{-(2\beta+1)}$, where a is the real-estimator variance
 63 constant and $cx^{-2\beta}$ is squared synthetic bias. This condition keeps the synthetic-variance term
 64 omitted by fixed-share shortcuts.
- 65 • **Phase diagram.** We characterize five allocation regimes indexed by the bias-decay exponent β
 66 and the synthetic-budget exponent ρ . In the headline regime, $\beta > 1/2$ and $0 < \rho < \beta + 1/2$, the
 67 optimal calibration size is $x_n^* \asymp n^{2\rho/(2\beta+1)}$, so the share $\lambda_n^* = x_n^*/n$ vanishes. When $\rho = 1$, this
 68 reduces to $x_n^* \asymp n^{2/(2\beta+1)}$: even linear synthetic scale calls for sublinear real-data calibration.
- 69 • **Adaptive implementation.** We give a grid estimator that learns the bias-decay curve on a held-out
 70 pilot grid, minimizes the corrected profiled risk over interior and boundary choices $x = 0, n$, and
 71 applies the safety check $x \cdot B_n(x) < a$ before using synthetic data. Under uniform consistency of
 72 the pilot risk estimates, the adaptive estimator matches the oracle in relative risk.
- 73 • **Inference and evidence.** We separate MSE-optimal point estimation from valid inference. The
 74 mixed estimator can reduce point risk while leaving enough residual synthetic bias to make
 75 naive Wald intervals undercover; validation debiasing restores coverage at a variance cost, in the
 76 spirit of double machine learning and prediction-powered inference [Chernozhukov et al., 2018,
 77 Angelopoulos et al., 2023]. Across roughly 10^7 Monte Carlo replications, the corrected oracle
 78 attains MSE 0.535 relative to all-real estimation in the headline cell, compared with 0.779 for a
 79 fixed-share baseline. In Experiment 4, validation-debiased intervals raise worst-cell coverage from
 80 0.855 to 0.938 at the nominal 0.95 level.

81 The rest of the paper proceeds as follows. Section 2 introduces the setup and notation. Section 3
 82 derives the optimal-mixing identity and corrected first-order condition. Section 4 gives the phase
 83 diagram. Section 5 develops the adaptive algorithm and inference results. Section 6 reports the
 84 simulations, and Section 7 concludes.

85 2 Setup

86 **Estimand and observables.** Fix a probability space and a measurable sample space $\mathcal{X}$. Let P^* be
 87 the real data-generating distribution. Let $\theta : \mathcal{P} \rightarrow \mathbb{R}$ be a scalar functional defined on a class $\mathcal{P}$ of
 88 distributions on $\mathcal{X}$ that contains P^* and the synthetic distributions introduced below. The target is
 89 $\theta^* := \theta(P^*)$. The researcher observes a real sample

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P^*,$$

90 where n is the total real-data budget. The researcher chooses an integer calibration budget $x \in$
 91 $\{0, 1, \dots, n\}$ and splits $[n] := \{1, \dots, n\}$ into a *calibration set* $C_x \subset [n]$ with $|C_x| = x$ and an
 92 *estimation set* $E_x := [n] \setminus C_x$ with $|E_x| = n_e := n - x$. The calibration share is $\lambda := x/n \in [0, 1]$.
 93 We enforce sample splitting throughout: estimators use only their assigned subsets, and the partition
 94 is independent of $\{X_i\}$. Let $\mathcal{F}_C := \sigma(\{X_i\}_{i \in C_x})$ denote the information in the calibration set.
 95 Intuitively, C_x is the data the generator learns from, and E_x is the data held back for direct estimation
 96 of θ^* .

97 **Generator and synthetic estimator.** A *calibration map* $\mathcal{G}$ takes the calibration sample $\{X_i\}_{i \in C_x}$
 98 and produces a calibrated synthetic distribution P_x . Conditionally on $\mathcal{F}_C$, draw synthetic observations

$$Z_1, \dots, Z_{m_n} \mid \mathcal{F}_C \stackrel{\text{iid}}{\sim} P_x,$$

99 where $m_n \in \mathbb{N}$ is the synthetic budget. The *pseudo-true* parameter is $\theta_x := \theta(P_x)$, which is $\mathcal{F}_C$ -
 100 measurable. The *synthetic bias* is $b(x) := \theta_x - \theta^*$, with second moment $b_2(x) := \mathbb{E}[(\theta_x - \theta^*)^2]$. In
 101 words, $b(x)$ measures how far the calibrated generator remains from the real target; spending more
 102 real data on calibration should make $b_2(x)$ smaller.

103 **Asymptotic linearity.** We require the real and synthetic estimators to be $\sqrt{|E_x|}$ - and $\sqrt{m_n}$ -
 104 asymptotically linear, respectively.

105 **Assumption 2.1** (Asymptotic linearity of the real estimator). There exists a measurable function
 106 $\psi : \mathcal{X} \rightarrow \mathbb{R}$ with $\mathbb{E}_{P^*}[\psi(X_1)] = 0$ and $\text{Var}_{P^*}(\psi(X_1)) = a \in (0, \infty)$ such that, for every
 107 deterministic sequence of estimation subsets $E \subset [n]$ with $|E| \rightarrow \infty$,

$$\hat{\theta}_R(E) - \theta^* = \frac{1}{|E|} \sum_{i \in E} \psi(X_i) + r_R(E),$$

108 where the remainder satisfies $\mathbb{E}[r_R(E)] = 0$ and $\mathbb{E}[r_R(E)^2] = o(|E|^{-1})$.

109 **Assumption 2.2** (Asymptotic linearity of the synthetic estimator). There exists a measurable family
 110 $(\phi_x)_{x \geq 1}$ such that, conditionally on $\mathcal{F}_C$,

$$\hat{\theta}_S(x) - \theta_x = \frac{1}{m_n} \sum_{j=1}^{m_n} \phi_x(Z_j) + r_S(x),$$

111 with $\mathbb{E}[\phi_x(Z_1) \mid \mathcal{F}_C] = 0$, $\text{Var}(\phi_x(Z_1) \mid \mathcal{F}_C) = \sigma_S^2(x)$, $\mathbb{E}[r_S(x) \mid \mathcal{F}_C] = 0$, and $\mathbb{E}[r_S(x)^2] =$
 112 $o(m_n^{-1})$. Furthermore there is a constant $\sigma_S^2 < \infty$ such that $\mathbb{E}[\sigma_S^2(x)] \rightarrow \sigma_S^2$ uniformly across the
 113 regimes considered.

114 The function ψ is the influence function of $\hat{\theta}_R$ and a is its variance. Assumption 2.1 delivers the
 115 leading-order real-estimator variance

$$V_R(x) := \text{Var}(\hat{\theta}_R(E_x)) = \frac{a}{n-x} \{1 + o(1)\},$$

116 while Assumption 2.2 gives $\text{Var}(\hat{\theta}_S(x) \mid \mathcal{F}_C) = \sigma_S^2/m_n + o(m_n^{-1})$. We will write $V_R(x) = a/(n-x)$
 117 and $\text{Var}(\hat{\theta}_S(x) \mid \mathcal{F}_C) = \sigma_S^2/m_n$ throughout, absorbing the $o(\cdot)$ remainders into the $1 + o(1)$ factors
 118 that multiply the leading-order risk.

119 **Working model.** The synthetic-error rate and the synthetic-variance rate are governed by two scalar
 120 exponents.

121 **Assumption 2.3** (Power-law calibration). There exist constants $c > 0$ and $\beta > 0$ such that

$$b_2(x) = c x^{-2\beta} \{1 + o(1)\} \quad \text{as } x \rightarrow \infty.$$

122 **Assumption 2.4** (Synthetic-variance regime). There exist constants $v_0 > 0$ and $\rho \geq 0$ such that the
 123 synthetic sampling variance $v_n := \sigma_S^2/m_n$ satisfies

$$v_n = v_0 n^{-\rho} \{1 + o(1)\} \quad \text{as } n \rightarrow \infty.$$

124 Assumption 2.3 encodes the empirical power-law bias decay often seen in neural scaling-law studies;
 125 β is the bias-decay exponent and c is the scale. Assumption 2.4 encodes how quickly the synthetic
 126 budget grows with the real budget; the main scenario $m_n \asymp n$ corresponds to $\rho = 1$. Combining the
 127 bias and variance pieces gives the *effective synthetic error*

$$B_n(x) := v_n + c x^{-2\beta},$$

128 which is, up to a $1 + o(1)$ factor, the unconditional MSE of $\hat{\theta}_S(x)$ around θ^* . The first term, v_n ,
 129 is Monte Carlo noise from drawing only m_n synthetic observations. The second term, $c x^{-2\beta}$, is
 130 residual generator bias after calibration. The allocation problem is to reduce $B_n(x)$ by increasing x ,
 131 while remembering that doing so leaves only $n - x$ real observations for direct estimation.

132 **Linear allocation estimator class.** For a mixing weight $\alpha \in [0, 1]$ and calibration budget $x \in [0, n]$,
 133 the *linear allocation estimator* is

$$\hat{\theta}^L(\alpha, x) := \alpha \hat{\theta}_R(E_x) + (1 - \alpha) \hat{\theta}_S(x). \quad (1)$$

134 In words, $\hat{\theta}^L$ is a convex combination of the real-only estimator on the held-out data and the synthetic-
 135 only estimator from the calibrated generator; the two corner cases $\alpha = 1$ and $\alpha = 0$ recover the
 136 all-real estimator on E_x and the synthetic-only estimator, respectively. Section 3 characterizes the
 137 optimal (α, x) pair under Assumptions 2.1, 2.2, 2.3, and 2.4.

138 3 Optimal mixing and corrected allocation

139 This section solves the allocation problem in two steps. First, Section 3.1 fixes the calibration budget
 140 x and chooses the best mixing weight α between the real and synthetic estimators. This gives a
 141 one-dimensional risk curve $R_n(x)$. Second, Section 3.2 differentiates that curve with respect to x
 142 and gives the corrected first-order condition. The difference from the usual heuristic is one retained
 143 term, the synthetic sampling variance v_n , and that term drives the phase diagram in Section 4. Finally,
 144 Section 3.3 gives a finite-sample safety check for falling back to all-real estimation. All proofs are
 145 deferred to Appendix C.

146 3.1 Optimal mixing for fixed allocation

147 We first compute the leading-order MSE of the linear allocation estimator and minimize it over the
 148 mixing weight. Throughout, $\hat{\theta}^L(\alpha, x)$ is as in Eq. (1), $V_R(x) = a/(n - x)$, and $B_n(x) = v_n + c x^{-2\beta}$.

149 **Lemma 3.1** (Working MSE). *Under Assumptions 2.1, 2.2 and 2.3,*

$$\mathbb{E}[(\hat{\theta}^L(\alpha, x) - \theta^*)^2] = \alpha^2 V_R(x) + (1 - \alpha)^2 B_n(x) + o(V_R(x)) + o(B_n(x)),$$

150 *uniformly in $\alpha \in [0, 1], x \in (0, n)$.*

151 In words, sample-splitting makes $\hat{\theta}_R(E_x)$ and $\hat{\theta}_S(x)$ unconditionally independent, so the cross term
 152 in the squared error vanishes and the MSE decomposes additively. The first term is the variance
 153 contribution from the real estimator, downweighted by α^2 ; the second is the unconditional MSE of
 154 the synthetic estimator around θ^* , downweighted by $(1 - \alpha)^2$.

155 Minimizing the leading-order MSE in α for fixed x is a one-dimensional convex problem and admits
 156 a closed-form solution.

157 **Proposition 3.2** (Optimal mixing weight and profiled risk). *Fix $x \in (0, n)$ such that $0 < B_n(x) < \infty$.*
 158 *Then the unique minimizer of $\alpha \mapsto \overline{\text{MSE}}_n(\alpha, x)$ over $\alpha \in [0, 1]$, where $\overline{\text{MSE}}_n(\alpha, x) := \alpha^2 V_R(x) +$*
 159 *$(1 - \alpha)^2 B_n(x)$, is*

$$\alpha^*(x) = \frac{B_n(x)}{V_R(x) + B_n(x)} \in (0, 1). \quad (2)$$

160 *The corresponding profiled risk is*

$$R_n(x) := \overline{\text{MSE}}_n(\alpha^*(x), x) = \frac{V_R(x)B_n(x)}{V_R(x) + B_n(x)} = \frac{aB_n(x)}{a + (n - x)B_n(x)}. \quad (3)$$

161 The formula has a simple reading. If $B_n(x)$ is large relative to $V_R(x)$, the synthetic estimator is
 162 unreliable, so the optimal rule puts most weight on real data ($\alpha^* \rightarrow 1$). If $B_n(x)$ is small, the
 163 synthetic estimator is accurate, so it gets most of the weight ($\alpha^* \rightarrow 0$). The profiled risk $R_n(x)$ is the
 164 usual inverse-variance pooling risk for two independent estimators, with $B_n(x)$ playing the role of
 165 the synthetic estimator's effective error. The remaining problem is to choose the calibration budget x
 166 that minimizes this profiled risk.

167 3.2 Exact corrected first-order condition

168 The fixed-share shortcut used as a baseline treats the synthetic estimator as if its only error were
 169 calibration bias. In our notation, it differentiates a risk surrogate after replacing $B_n(x) = v_n + cx^{-2\beta}$
 170 by $cx^{-2\beta}$. Dropping v_n is harmless only when synthetic sampling variance is negligible at the
 171 relevant allocation. It is not harmless when the synthetic budget m_n grows with n , because v_n then
 172 enters the marginal value of one more calibration observation. Keeping this term produces a different
 173 first-order condition, one that leads to the phase transition in Section 4 rather than a rate-independent
 174 fixed-share rule such as $\lambda = 2\beta/(1 + 2\beta)$.

175 **Proposition 3.3** (Exact first-order condition). *Under Assumption 2.3 and 2.4 with $v_n > 0$, every*
 176 *interior critical point $x \in (0, n)$ of R_n satisfies*

$$(v_n + cx^{-2\beta})^2 = 2\beta acx^{-(2\beta+1)}, \quad (4)$$

177 *or equivalently, after multiplying through by $x^{2\beta+1}$,*

$$v_n^2 x^{2\beta+1} + 2cv_n x + c^2 x^{1-2\beta} = 2\beta ac. \quad (5)$$

178 *For $\beta \leq 1/2$ this equation has at most one solution on $(0, \infty)$; for $\beta > 1/2$ it has at most two, and*
 179 *the active interior root is the larger.*

180 The left-hand side of Eq. (4) is $B_n(x)^2$, the squared effective synthetic error. The right-hand side is
 181 the cost, in real-estimator precision, of spending one more real observation on calibration. Thus the
 182 optimum balances synthetic-error reduction against real-data loss. The mixed term $2cv_n x$ in Eq. (5)
 183 is the piece lost when one drops synthetic sampling variance; keeping it is the algebraic content of the
 184 correction. The root structure is controlled by the auxiliary function $\Phi(x) := B_n(x)^2 x^{2\beta+1}$, which
 185 is monotone for $\beta \leq 1/2$ and U-shaped for $\beta > 1/2$. Section 4 turns that shape into the allocation
 186 phase diagram.

187 3.3 Safe-improvement condition

188 The corrected FOC characterizes interior optima of R_n , but the global oracle is taken over the closed
 189 interval $[0, n]$, so the boundary $x = 0$ (no calibration, all-real estimator on n observations, with risk
 190 a/n) must be compared against any interior candidate. The next result reduces this comparison to a
 191 one-line algebraic check.

192 **Proposition 3.4** (Exact safe-improvement condition). *For every $x \in (0, n)$,*

$$R_n(x) < \frac{a}{n} \iff xB_n(x) < a. \quad (6)$$

193 *Equivalently, $x(v_n + cx^{-2\beta}) < a$. The boundary risk is $R_n(0) = a/n$.*

In words, calibration budget x beats all-real estimation on n observations exactly when $x B_n(x) < a$. The product $x B_n(x)$ compares the amount of real data spent on calibration with the remaining error of the calibrated synthetic estimator. If the inequality fails because v_n is too large or because the bias term $cx^{-2\beta}$ would require too much calibration to shrink, the procedure returns the all-real estimator, $\hat{x} = 0$, $\hat{\alpha} = 1$. Section 3.2 produces candidate interior optima; Eq. (6) decides whether any candidate should be used. The adaptive procedure in Section 5 minimizes $R_n(x)$ on a grid that includes $x = 0$ and $x = n$, then applies this safety check.

4 Allocation phase diagram

The corrected first-order condition from Section 3 is one algebraic equation, but its solution can fall in different parts of the feasible interval $[0, n]$. Which case occurs depends on two exponents: the generator bias-decay exponent β and the synthetic-variance exponent ρ from Assumption 2.3 and 2.4. This section maps those cases. Section 4.1 studies the shape of the risk curve, Section 4.2 states the five-regime phase diagram, and Section 4.3 gives the risk rates; Section A explains what changes when the generator has irreducible bias.

4.1 Critical-point structure

Multiplying both sides of the first-order condition Eq. (4) by $x^{2\beta+1}$ removes negative powers of x and leaves one object to study:

$$\Phi(x) := (v_n + cx^{-2\beta})^2 x^{2\beta+1} = B_n(x)^2 x^{2\beta+1}, \quad (7)$$

The function Φ is monotone when learning is slow and U-shaped when learning is fast. Critical points of $R_n(x)$ are exactly the roots of $\Phi(x) = 2\beta ac$. Thus the number of candidate optima is determined by the shape of this one curve.

Lemma 4.1 (Critical points of R_n). *Assume Assumption 2.3 and 2.4 with $a, c, v_n > 0$.*

(i) *If $\beta \leq 1/2$, then Φ is strictly increasing on $(0, \infty)$, so R_n has at most one interior critical point on $(0, n)$, and when it exists it is a strict local minimum. For $\beta < 1/2$ the critical point exists for every $v_n > 0$; for $\beta = 1/2$ it exists if and only if $a > c$.*

(ii) *If $\beta > 1/2$, then Φ is strictly U-shaped with unique minimum at*

$$x^\dagger = \left(\frac{c(2\beta-1)}{(2\beta+1)v_n} \right)^{1/(2\beta)}, \quad \Phi^\dagger := \Phi(x^\dagger) \propto c^{(2\beta+1)/(2\beta)} v_n^{(2\beta-1)/(2\beta)}. \quad (8)$$

When $\Phi^\dagger < 2\beta ac$, R_n has exactly two interior critical points $x_1 < x^\dagger < x_2$; the smaller is a local maximum and the larger, $x_n^\circ := x_2$, is the unique interior local minimum. When $\Phi^\dagger \geq 2\beta ac$, R_n has no interior local minimum. Under Assumption 2.4 with $\rho > 0$, $\Phi^\dagger \rightarrow 0$, so the two-root regime holds for all sufficiently large n .

That is to say, when the generator learns slowly ($\beta \leq 1/2$), the bias term $cx^{-2\beta}$ decays gently and the profiled risk has at most one interior minimum. When the generator learns quickly ($\beta > 1/2$), the bias decays steeply enough that Φ first falls and then rises. The active minimum is the larger of the two roots; the smaller root is a local maximum and is not useful for allocation. The value $\Phi^\dagger$ tells us whether the dip falls below $2\beta ac$. Once n is large enough, it does, so the active root x_n° exists.

4.2 Phase diagram

The asymptotic location of the active root x_n° , and hence of the global oracle $x_n^* \in \arg \min_{x \in [0, n]} R_n(x)$, depends on (β, ρ) in five distinct ways.

Theorem 4.2 (Allocation phase diagram). *Assume Assumption 2.3 and 2.4. Let x_n° denote the active interior root from Theorem 4.1 (when it exists) and $x_n^* \in \arg \min_{x \in [0, n]} R_n(x)$ the global oracle. As $n \rightarrow \infty$:*

Regime I (persistent synthetic variance, $\rho = 0$). *Every interior critical point is $O(1)$, so $\lambda_n^* = x_n^*/n \rightarrow 0$ regardless of β .*

236 **Regime II (slow learning, $\beta < 1/2, \rho > 0$).** $x_n^\circ \rightarrow x_\infty := (2\beta a/c)^{1/(1-2\beta)} \in (0, \infty)$, hence
 237 $x_n^\circ = O(1)$ and $\lambda_n^* \rightarrow 0$. The first-order rate of $R_n(x_n^*)$ matches the all-real rate a/n , with strict
 238 finite-sample shrinkage.

239 **Regime III (sublinear calibration, $\beta > 1/2, 0 < \rho < \beta + \frac{1}{2}$).** For all sufficiently large n , the active
 240 interior root exists and satisfies

$$x_n^\circ = \left(\frac{2\beta ac}{v_0^2} \right)^{1/(2\beta+1)} n^{2\rho/(2\beta+1)} \{1 + o(1)\}. \quad (9)$$

241 Consequently $\lambda_n^\circ = x_n^\circ/n \asymp n^{2\rho/(2\beta+1)-1} \rightarrow 0$, and the safe-improvement condition of Theorem 3.4
 242 holds asymptotically: $x_n^\circ B_n(x_n^\circ) \rightarrow 0 < a$. Hence $x_n^* = x_n^\circ \{1 + o(1)\}$.

243 **Regime IV (full calibration, $\beta > 1/2, \rho > \beta + \frac{1}{2}$).** The unconstrained Regime III root would exceed
 244 the real-data budget: $x_n^\circ \gg n$. The constrained oracle therefore pins to the boundary $x = n$, where
 245 $B_n(n) = v_n + cn^{-2\beta} = o(n^{-1})$. Thus $\lambda_n^* \rightarrow 1$ and $R_n(x_n^*) = v_n + cn^{-2\beta} = o(n^{-1})$, beating the
 246 all-real rate.

247 **Regime V (parametric knife-edge, $\beta = 1/2$).** An interior root exists iff $a > c$; in that case
 248 $x_n^\circ = (\sqrt{ac} - c) n^\rho / v_0 \{1 + o(1)\}$. The boundary regime depends on ρ and on the constants
 249 (a, c, v_0) : for $\rho < 1$, $\lambda_n^\circ \rightarrow 0$ with $R_n(x_n^*) = a/n \{1 + o(1)\}$ and a strict finite-sample improvement;
 250 for $\rho = 1$, $\lambda_n^\circ \rightarrow (\sqrt{ac} - c)/v_0 \in (0, 1)$ provided $\sqrt{ac} - c < v_0$; for $\rho > 1$, $\lambda_n^* \rightarrow 1$ if $c < a$. If
 251 $a \leq c$, no interior root exists and $x_n^* = 0$.

252 Figure 1 shades the (β, ρ) plane by these five regimes. The main case is the row $\rho = 1, \beta > 1/2$
 253 inside Regime III. When synthetic and real budgets scale together and the generator learns faster
 254 than parametric, Eq. (9) gives $x_n^* \asymp n^{2/(2\beta+1)}$. The optimal calibration size grows, but sublinearly,
 255 so the calibration share $\lambda_n^* \rightarrow 0$. This explains why positive fixed-share baselines, including
 256 $\lambda = 2\beta/(1 + 2\beta)$, over-calibrate in the common regime $m_n \asymp n$.

257 4.3 Risk consequences

258 Inside Regime III the variance side $V_R(x_n^\circ) \sim a/n$ and bias side $B_n(x_n^\circ) \sim v_n$ collapse the harmonic-
 259 mean profiled risk into one of three subcases driven by whether the synthetic budget grows slower
 260 than, in step with, or faster than the real budget:

$$R_n(x_n^*) = \frac{V_R(x_n^\circ) v_n}{V_R(x_n^\circ) + v_n} \{1 + o(1)\} = \begin{cases} \frac{a}{n} \{1 + o(1)\} & \text{if } \rho < 1, \\ \frac{av_0}{(a + v_0)n} \{1 + o(1)\} & \text{if } \rho = 1, \\ \frac{v_0}{n^\rho} \{1 + o(1)\} & \text{if } \rho > 1. \end{cases} \quad (10)$$

261 The three branches have a simple interpretation. If $\rho < 1$, synthetic sampling noise is too large to
 262 change the first-order rate, so the risk matches all-real up to a smaller-order gain. If $\rho = 1$, real and
 263 synthetic variances are both order $1/n$, and pooling gives a strictly smaller leading constant than a/n .
 264 If $\rho > 1$, the synthetic side is accurate enough that the risk is faster than $1/n$, so all-real estimation is
 265 first-order suboptimal.

266 5 Adaptive method and inference

267 The oracle allocation x_n^* in Theorem 4.2 depends on unknown quantities: the real-estimator variance
 268 constant a , the synthetic-variance scale v_n , and the bias-curve constants (c, β) . Section 5.1 gives a
 269 grid estimator that learns these quantities from pilot calibrations. Section 5.2 shows that it tracks
 270 the oracle when the estimated risk curve is uniformly accurate in relative terms. Section B (in
 271 the appendix, for space) then explains when the resulting point estimator supports ordinary Wald
 272 intervals.

273 5.1 Adaptive grid-minimization estimator

274 The idea is to estimate the profiled risk $R_n(x)$ on a discrete grid, choose the smallest estimated risk,
 275 and then apply the exact safety check $x B_n(x) < a$ from Theorem 3.4. We use grid minimization

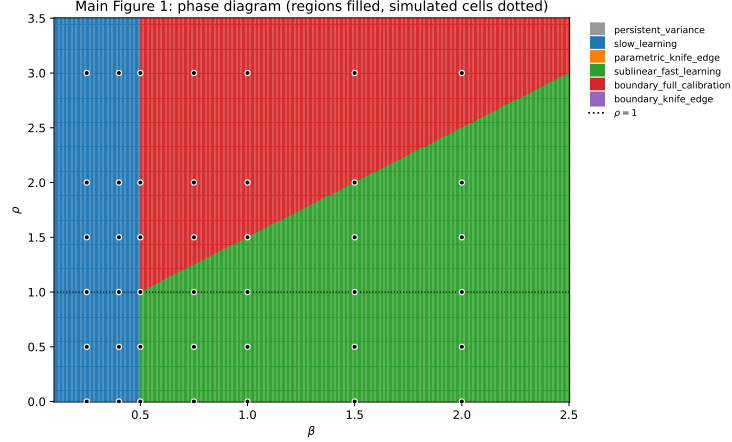


Figure 1: Theoretical phase diagram in (β, ρ) space. The five regimes of Theorem 4.2 partition the plane by the asymptotic behavior of the optimal calibration size x_n^* . The horizontal line $\rho = 1$ is the empirically common case in which the synthetic budget scales linearly with the real budget; the vertical line $\beta = 1/2$ is the parametric knife edge. The labelled headline regime is the wide wedge $\beta > 1/2, 0 < \rho < \beta + \frac{1}{2}$, where the optimal calibration share vanishes despite the optimal calibration size diverging.

rather than solving the first-order condition Eq. (4) directly. The grid handles the boundary choices $x = 0$ and $x = n$, avoids the smaller local maximum from Theorem 4.1, and does not require estimating β at a parametric rate.

The full pseudocode (Algorithm 1 in Section D) uses the safe-improvement boundary from Theorem 3.4 directly. The same inequality that defines when synthetic augmentation can help also tells the procedure when to abstain. In step 6 of the pseudocode the squared bias is validation-debiased as $\hat{b}^2(x_j) = [(\hat{\theta}_S(x_j) - \hat{\theta}_R^V)^2 - \widehat{\text{Var}}(\hat{\theta}_S(x_j)) - \widehat{\text{Var}}(\hat{\theta}_R^V)]_+$, and the truncation $[\cdot]_+$ prevents Monte Carlo noise from producing negative squared-bias estimates; below we assume that this truncation is harmless on the part of the grid where variance dominates.

5.2 Oracle inequality

Grid minimization does not require $\hat{R}_n$ to be accurate in absolute units. It only needs to rank grid points correctly with high probability. The right condition is uniform relative consistency.

Theorem 5.1 (Adaptive oracle inequality). *Let $\mathcal{G}_n^+ := \mathcal{G}_n \cup \{0, n\}$, $x_n^* := \arg \min_{x \in \mathcal{G}_n^+} R_n(x)$, and $\hat{x}_n := \arg \min_{x \in \mathcal{G}_n^+} \hat{R}_n(x)$. Suppose*

$$\Delta_n := \sup_{x \in \mathcal{G}_n^+} \left| \frac{\hat{R}_n(x)}{R_n(x)} - 1 \right| \xrightarrow{P} 0. \quad (11)$$

Then $R_n(\hat{x}_n)/R_n(x_n^) \xrightarrow{P} 1$. If furthermore the grid satisfies $\inf_{x \in \mathcal{G}_n} R_n(x)/\inf_{x \in [0, n]} R_n(x) \rightarrow 1$, then $R_n(\hat{x}_n)/\inf_{x \in [0, n]} R_n(x) \xrightarrow{P} 1$.*

In words, the empirical risk curve only needs to be uniformly close to the true curve in *ratio*. A constant multiplicative error in $\hat{R}_n$ does not change the argmin, so the grid procedure can inherit the oracle rate even when the individual estimates of a , v_n , c , and β converge slowly. This is why the grid version is preferable to a plug-in solver of the first-order condition, which would be more sensitive to slow convergence of $\hat{\beta}$.

6 Experiments

The experiments answer three questions. First, does the corrected oracle obey the phase diagram? Second, can an adaptive estimator recover useful allocations from finite pilot data? Third, when does

the MSE-optimal estimator need validation debiasing for inference? Experiments 1, 2, and 4 use full synthetic Monte Carlo sweeps; Experiments 1 and 2 together use roughly 10^7 replications, and Experiment 4 uses 280,000 replications. The multichannel, tabular, and causal benchmarks, together with complete result tables for all experiments, are reported in Appendix E.

6.1 Phase-diagram validation

Setup. Real observations follow $X_i = \theta^* + \epsilon_i$ with $\epsilon_i \sim N(0, a)$. After calibration size x , synthetic observations follow $Z_j(x) = \theta^* + B_0 x^{-\beta} + u_j$ with $u_j \sim N(0, \sigma_S^2)$ and $m_n = \lceil \kappa n^\rho \rceil$. The main grid sets $\theta^* = 0$ and $a = B_0 = c = \sigma_S^2 = \kappa = 1$, then sweeps $n \in \{100, \dots, 20,000\}$, seven values of $\beta \in [0.25, 2]$, and six values of $\rho \in [0, 3]$.

Main evidence. The oracle results support the paper’s main claim. In the headline cell ($\beta = 1$, $\rho = 1$, $n = 20,000$), the corrected oracle has MSE ratio 0.535 relative to all-real estimation. The old fixed-share rule has ratio 0.779, and synthetic-only full calibration has ratio 1.015. Thus the corrected allocation roughly doubles the MSE reduction obtained by the fixed-share rule, while synthetic-only use is slightly harmful. Figure 3 shows the same ordering across the grid. The scaling check in Figure 2 also matches the theory: among the interior fast-learning cells, the four disagreements are all upward deviations near the slow-learning boundary, consistent with finite-grid pre-asymptotics rather than a failure of the first-order condition. Full slope estimates are in Table 1.

6.2 Adaptive feasibility

Experiment 2 uses the same Gaussian DGP on a smaller grid and estimates the bias curve from the pilot grid $\mathcal{G}_{\text{pilot}} = \{5, 10, 20, 40, 80, 160, 320, 640\} \cap [1, n/2]$. The adaptive story is more cautious than the oracle story. In the headline cell ($\beta = 1$, $\rho = 1$, $n = 10,000$), the oracle MSE ratio is 0.522, while `corrected_adaptive_gn` reaches 0.867 and `safe_corrected_adaptive_gn` reaches 0.908. The selected calibration size is close to oracle in that cell (563 versus 573), but the MSE gap shows that allocation tracking is not the same as oracle-equivalent point estimation.

Across the 60 non-trivial cells with $\rho > 0$, `corrected_adaptive_gn` is harmful in 21 cells, with median MSE ratio 0.986 and maximum ratio 1.460. This is not evidence that the adaptive method has reached its asymptotic oracle regime at the simulated sample sizes. It is evidence for a more modest claim: the corrected risk curve and safety fallback make adaptive synthetic augmentation stable enough to avoid the much larger failures of naive synthetic-use rules. Figure 4a and Figure 4b show allocation tracking and residual regret; full adaptive summaries are in Table 2.

6.3 Inference and coverage

Experiment 4 focuses on regimes where residual synthetic bias can be large relative to standard error: $\beta \in \{0.4, 0.5, 0.75, 1.0\}$, $\rho = 1$, and $n \in \{200, 500, 1000, 2000, 5000\}$. The point-estimation message and the inference message separate cleanly. Naive Wald intervals around the MSE-optimal estimator have the weakest coverage, with full-grid range 0.855–0.955 and median 0.916. Validation-debiased intervals have range 0.938–0.964 and median 0.953, closest to nominal among the synthetic intervals. The worst naive cell is the parametric knife edge ($\beta = 0.5$, $\rho = 1$, $n = 500$), where naive coverage is 0.855 and validation-debiased coverage is 0.954. Figure 5 plots the full coverage curves, and Table 3 gives the complete coverage summary.

Reproducibility. Full configurations, raw and aggregated long-format results, and the analysis notebook used to render Figure 2–Figure 5 are available at <anon> (URL withheld for blind submission).

7 Conclusion

Synthetic data do not eliminate the value of real data. They make real data more strategic. The paper’s main message is that scarce real observations have two competing roles: they are evidence for the estimand and inputs that improve the synthetic generator. Treating calibration as a fixed-share preprocessing step misses this trade-off. The correct allocation is governed by marginal value: one more real observation should be spent on calibration only when the reduction in synthetic error is worth the loss of direct real-estimation precision.

This marginal-value logic changes the practical recommendation. In the central regime for modern synthetic-data pipelines, where synthetic samples scale with the real-data budget and generator bias declines quickly, the optimal calibration size grows but the calibration share vanishes. More synthetic scale does not mean spending a constant fraction of real data on calibration. It means using a growing, sublinear calibration set and preserving most real observations for direct estimation.

The same distinction matters for uncertainty quantification. MSE-optimal synthetic augmentation can improve point estimation while leaving residual bias large enough to damage naive confidence intervals. Valid inference therefore requires an explicit bias correction or an allocation rule designed for coverage rather than point risk. The broader lesson is simple: synthetic data are useful only when they are allocated, weighted, and debiased with the real-data budget in view.

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A Plateau bias

This subsection was moved from the main paper (formerly Section 4.4) to fit the 9-page content limit; it is reproduced here verbatim.

The phase diagram assumes the synthetic squared bias drives to zero with calibration. Foundation-model-style generators may instead leave a non-vanishing bias floor, such as alignment errors that more calibration data cannot remove. The next result records that, once such a floor is present, the regime structure collapses.

Proposition A.1 (Plateau-bias regime). *Replace Assumption 2.3 by $b_2(x) = b_\infty^2 + cx^{-2\beta}\{1 + o(1)\}$ with $b_\infty^2 > 0$ and $\beta > 0$, and suppose Assumption 2.4 holds with $\rho \geq 0$. Then for n sufficiently large the active interior root satisfies*

$$x_n^\circ \leq \left(\frac{2\beta ac}{b_\infty^4} \right)^{1/(2\beta+1)} = O(1), \quad (12)$$

so $\lambda_n^\circ \rightarrow 0$ for every $\beta > 0$ and every $\rho \geq 0$. The optimal risk satisfies $R_n(x_n^\star) = a/n\{1 + o(1)\}$: synthetic augmentation buys at most a finite-sample shrinkage gain and no first-order rate improvement. The shrinkage is positive iff $x_n^\circ b_\infty^2 < a$.

In words, plateau bias caps calibration at a constant budget regardless of how fast the bias *would* have decayed before reaching the floor. Operationally, the generator is useful only down to its noise floor; beyond that point, further calibration is wasted.

B Asymptotic distribution and Wald inference

This subsection was moved from the main paper (formerly Section 5.3) to fit the 9-page content limit; it is reproduced here verbatim.

A low-MSE point estimator does not automatically give a valid confidence interval. The same residual bias that the corrected oracle tolerates in mean square appears as an additive shift in the centered limit law. The next two results make that shift explicit.

Proposition B.1 (Centered asymptotic normality). *Under Assumptions 2.1, 2.2, 2.3, and 2.4 with $v_n > 0$ and a conditional Lindeberg condition on ψ and ϕ_x , the linear estimator at the oracle satisfies*

$$\frac{\hat{\theta}^L(\alpha^*(x_n^\star), x_n^\star) - \theta^* - (1 - \alpha^*(x_n^\star)) \Delta(x_n^\star)}{\sqrt{\Omega_n(x_n^\star)}} \Rightarrow N(0, 1), \quad (13)$$

conditionally on $\mathcal{F}_C$ and unconditionally, where $\Omega_n(x) := \alpha^(x)^2 V_R(x) + (1 - \alpha^*(x))^2 v_n$ is the stochastic-variance contribution and $\Delta(x) := \theta_x - \theta^*$ is the calibration-side bias. Replacing oracle quantities by their estimates from Theorem 5.1 preserves the CLT.*

In plain English, the estimator is asymptotically normal around a shifted center, $\theta^* + (1 - \alpha^*(x_n^\star)) \Delta(x_n^\star)$, not necessarily around θ^* . The shift is the synthetic weight times the remaining calibration bias. It matters for inference only when it is large relative to the standard error.

Corollary B.2 (Wald inference around θ^*). *A two-sided Wald interval centered at $\hat{\theta}^L$ with width $z_{1-\alpha/2} \sqrt{\Omega_n(x_n^\star)}$ has asymptotic coverage $1 - \alpha$ for θ^* if and only if*

$$(1 - \alpha^*(x_n^\star)) \Delta(x_n^\star) = o_p(\sqrt{\Omega_n(x_n^\star)}). \quad (14)$$

Under Theorem 4.2, in Regime III with $\rho = 1$ and $\beta > 1/2$, condition Eq. 14 holds; in Regime V with $\rho = 1$ and $\beta = 1/2$ it generically fails.

In Regime III, calibration is fast enough that $\Delta(x_n^\star)$ is of order $n^{-2\beta/(2\beta+1)} = o(n^{-1/2})$, smaller than the standard error $\sqrt{\Omega_n} \asymp n^{-1/2}$. Naive Wald intervals are therefore asymptotically valid in that regime, though finite samples can still show undercoverage. In the parametric knife-edge Regime V, $\Delta(x_n^\star)$ and $\sqrt{\Omega_n}$ are the same order, and the shift in Eq. 13 produces undercoverage that does not vanish with n . In such cases, the validation-debiased construction of a PPI-type estimator subtracts an estimate of $\Delta(x_n^\star)$ using the held-out validation slice and restores nominal coverage at the cost of a constant-factor variance penalty.

C Proofs of main results

This appendix collects full proofs of the formal statements in the main paper. Throughout, we work in the linear-allocation configuration of Section 2: the index set $[n] = \{1, \dots, n\}$ is partitioned into the calibration block C_x of size x and the estimation block $E_x = [n] \setminus C_x$ of size $n - x$, and we write $V_R(x) = a/(n - x)$ for the variance of the real-only estimator $\hat{\theta}_R$ on E_x and $B_n(x) = v_n + cx^{-2\beta}$ for the effective synthetic-error variance. Assumptions 2.1, 2.2, 2.3, and 2.4 are in force throughout. The notation matches the main paper, including the convention $B_n(\cdot)$ for the effective synthetic-error variance.

C.1 Auxiliary lemmas

We use two auxiliary lemmas repeatedly. Both follow from sample splitting and the working model and are standard.

Lemma C.1 (Conditional and unconditional independence). *Fix x and suppose the split (C_x, E_x) is independent of $\{X_i\}_{i=1}^n$. Assume that, conditionally on $\mathcal{F}_C$, the synthetic observations $Z_1, \dots, Z_{m_n}$ are generated from an auxiliary random stream independent of $\{X_i : i \in E_x\}$. Then:*

$$\hat{\theta}_R(E_x) \perp\!\!\!\perp \hat{\theta}_S(x) \mid \mathcal{F}_C.$$

Moreover,

$$\hat{\theta}_R(E_x) \perp\!\!\!\perp (\theta_x, \hat{\theta}_S(x)).$$

Consequently, $\hat{\theta}_R(E_x)$ is independent of both $\hat{\theta}_S(x) - \theta_x$ and $\hat{\theta}_S(x) - \theta^*$.

Proof. Because the split is independent of the data and the observations are i.i.d., the estimation sample $\{X_i : i \in E_x\}$ is independent of $\mathcal{F}_C$. Conditional on $\mathcal{F}_C$, the calibrated distribution P_x and the pseudo-true parameter $\theta_x = \theta(P_x)$ are fixed, and the synthetic sample $Z_1, \dots, Z_{m_n}$ is generated using an auxiliary random stream independent of the estimation sample. Hence $\hat{\theta}_R(E_x)$ and $\hat{\theta}_S(x)$ are conditionally independent given $\mathcal{F}_C$.

For the unconditional statement, $\hat{\theta}_R(E_x)$ is measurable with respect to the estimation sample, while $(\theta_x, \hat{\theta}_S(x))$ is measurable with respect to the calibration sample together with the auxiliary synthetic stream. These two sources of randomness are independent. Therefore

$$\hat{\theta}_R(E_x) \perp\!\!\!\perp (\theta_x, \hat{\theta}_S(x)).$$

The two displayed consequences follow because $\hat{\theta}_S(x) - \theta_x$ and $\hat{\theta}_S(x) - \theta^*$ are measurable functions of $(\theta_x, \hat{\theta}_S(x))$. $\square$

Lemma C.2 (Synthetic-error decomposition). *Under Assumption 2.2 and 2.3, the synthetic estimator's mean-squared error around θ^* admits the decomposition*

$$\mathbb{E}[(\hat{\theta}_S(x) - \theta^*)^2] = v_n + b_2(x) + o(v_n) + o(b_2(x)) = B_n(x)\{1 + o(1)\},$$

where $b(x) := \theta_x - \theta^*$ is the calibration-induced bias and $B_n(x) := v_n + cx^{-2\beta}$.

Proof of Theorem C.2. Decompose $\hat{\theta}_S(x) - \theta^* = (\hat{\theta}_S(x) - \theta_x) + b(x)$. By Assumption 2.2, the first term has conditional mean zero and conditional variance $\sigma_S^2/m_n + o_p(1/m_n) = v_n\{1 + o(1)\}$ (using $v_n = \sigma_S^2/m_n$ from the working model). Squaring and taking the unconditional expectation, the cross term vanishes (mean-zero conditional on the calibration data, applied with the tower property), and the squared bias term equals $b_2(x) = cx^{-2\beta}\{1 + o(1)\}$ by Assumption 2.3. Combining gives the displayed decomposition. $\square$

C.2 Proof of Theorem 3.1 (Working MSE)

Proof of Theorem 3.1. Fix x in the calibration range under consideration and write $\xi_R := \hat{\theta}_R(E_x) - \theta^*$ and $\xi_S := \hat{\theta}_S(x) - \theta^*$. Then $\hat{\theta}^L(\alpha, x) - \theta^* = \alpha\xi_R + (1 - \alpha)\xi_S$. By the unconditional independence statement in Lemma C.1, ξ_R and ξ_S are independent. Hence

$$\mathbb{E}[(\alpha\xi_R + (1 - \alpha)\xi_S)^2] = \alpha^2\mathbb{E}[\xi_R^2] + (1 - \alpha)^2\mathbb{E}[\xi_S^2] + 2\alpha(1 - \alpha)\mathbb{E}[\xi_R]\mathbb{E}[\xi_S].$$

By Assumption 2.1, the real estimator has a mean-zero influence function and a mean-zero remainder, so $\mathbb{E}[\xi_R] = 0$. Thus the cross term vanishes exactly. Moreover, Assumption 2.1 gives $\mathbb{E}[\xi_R^2] = V_R(x)\{1 + o(1)\}$, uniformly over the calibration range. By Theorem C.2, the synthetic-error decomposition gives $\mathbb{E}[\xi_S^2] = B_n(x)\{1 + o(1)\}$, uniformly over the same range. Substituting these two expansions into the preceding display yields

$$\mathbb{E}[(\hat{\theta}^L(\alpha, x) - \theta^*)^2] = \alpha^2 V_R(x) + (1 - \alpha)^2 B_n(x) + \mathcal{R}_n(\alpha, x),$$

where $\mathcal{R}_n(\alpha, x) = \alpha^2 o\{V_R(x)\} + (1 - \alpha)^2 o\{B_n(x)\}$. Since $\alpha \in [0, 1]$, both α^2 and $(1 - \alpha)^2$ are bounded by one, and therefore $\mathcal{R}_n(\alpha, x) = o\{V_R(x)\} + o\{B_n(x)\}$ uniformly in $\alpha \in [0, 1]$. This proves the claim. $\square$

C.3 Proof of Theorem 3.2 (Optimal mixing)

Proof of Theorem 3.2. Define the leading-order MSE

$$\overline{\text{MSE}}_n(\alpha, x) := \alpha^2 V_R(x) + (1 - \alpha)^2 B_n(x).$$

This is a quadratic function of α . Its second derivative with respect to α is $2[V_R(x) + B_n(x)] > 0$ since both terms are strictly positive, so $\overline{\text{MSE}}_n(\cdot, x)$ is strictly convex. Setting the first derivative to zero gives

$$\frac{\partial}{\partial \alpha} \overline{\text{MSE}}_n(\alpha, x) = 2\alpha V_R(x) - 2(1 - \alpha)B_n(x) = 0,$$

which solves to $\alpha^*(x) = B_n(x)/[V_R(x) + B_n(x)] \in (0, 1)$ since both $V_R(x)$ and $B_n(x)$ are strictly positive and finite. This is the unique unconstrained minimizer; because it lies in the open interval $(0, 1)$, it is also the unique minimizer over $[0, 1]$.

Substituting $\alpha = \alpha^*(x)$ back into $\overline{\text{MSE}}_n$ and simplifying,

$$R_n(x) := \overline{\text{MSE}}_n(\alpha^*(x), x) = \frac{B_n(x)^2}{[V_R(x) + B_n(x)]^2} V_R(x) + \frac{V_R(x)^2}{[V_R(x) + B_n(x)]^2} B_n(x) = \frac{V_R(x)B_n(x)}{V_R(x) + B_n(x)},$$

which is the harmonic-mean form. Multiplying numerator and denominator by $(n - x)$ and using $V_R(x) = a/(n - x)$ gives the second equality

$$R_n(x) = \frac{a B_n(x)}{a + (n - x) B_n(x)}. \quad \square$$

C.4 Proof of Theorem 3.3 (Exact FOC)

To prove Theorem 3.3 we first record an intermediate result on the gradient of the profiled risk. This auxiliary lemma is local to the appendix.

Define

$$\Phi(x) := B_n(x)^2 x^{2\beta+1}, \quad x > 0, \quad (15)$$

and

$$H(x) := a B'_n(x) + B_n(x)^2. \quad (16)$$

Lemma C.3 (Gradient of the profiled risk). *For every $x \in (0, n)$,*

$$R'_n(x) = \frac{a H(x)}{[a + (n - x) B_n(x)]^2} = \frac{a}{[a + (n - x) B_n(x)]^2} \cdot \frac{\Phi(x) - 2\beta a c}{x^{2\beta+1}}. \quad (17)$$

In particular, $\text{sign}(R'_n(x)) = \text{sign}(\Phi(x) - 2\beta a c)$.

Proof. Write $R_n = u/w$ with $u(x) = a B_n(x)$ and $w(x) = a + (n - x) B_n(x)$. Differentiating,

$$u'(x) = a B'_n(x), \quad w'(x) = -B_n(x) + (n - x) B'_n(x).$$

By the quotient rule, $R'_n(x) = (u'w - uw')/w^2$. Compute the numerator:

$$\begin{aligned} u'w - uw' &= a B'_n [a + (n - x) B_n] - a B_n [-B_n + (n - x) B'_n] \\ &= a^2 B'_n + a(n - x) B_n B'_n + a B_n^2 - a(n - x) B_n B'_n \\ &= a^2 B'_n + a B_n^2 = a H(x). \end{aligned}$$

497 The two cross terms $\pm a(n-x)B_n B'_n$ cancel exactly; this algebraic cancellation is the reason the
 498 optimum does not depend on $n-x$ at first order.

499 To rewrite in terms of Φ , substitute $B'_n(x) = -2\beta c x^{-(2\beta+1)}$ (from $B_n(x) = v_n + cx^{-2\beta}$) and
 500 multiply $H(x)$ by $x^{2\beta+1}$:

$$H(x)x^{2\beta+1} = -2\beta ac + B_n(x)^2 x^{2\beta+1} = \Phi(x) - 2\beta ac.$$

501 Dividing by $x^{2\beta+1}$ and substituting into $R'_n(x) = aH(x)/w^2$ gives the second equality in Eq. (17).
 502 The denominator $[a + (n-x)B_n(x)]^2$ is strictly positive on $(0, n)$, and a and $x^{2\beta+1}$ are strictly
 503 positive, so $R'_n(x)$ has the same sign as $\Phi(x) - 2\beta ac$. $\square$

504 *Proof of Theorem 3.3.* By Theorem C.3, the interior critical points of R_n on $(0, n)$ are exactly the
 505 points where $H(x) = 0$, equivalently $\Phi(x) = 2\beta ac$. Writing $\Phi(x) = (v_n + cx^{-2\beta})^2 x^{2\beta+1}$ and
 506 dividing both sides of $\Phi(x) = 2\beta ac$ by $x^{2\beta+1}$ gives the FOC

$$(v_n + cx^{-2\beta})^2 = 2\beta ac x^{-(2\beta+1)},$$

507 which is Eq. (4) of the main paper. Multiplying by $x^{2\beta+1}$ and expanding the square,

$$(v_n + cx^{-2\beta})^2 x^{2\beta+1} = v_n^2 x^{2\beta+1} + 2v_n cx + c^2 x^{1-2\beta},$$

508 so the polynomial form

$$v_n^2 x^{2\beta+1} + 2v_n cx + c^2 x^{1-2\beta} = 2\beta ac$$

509 is equivalent. The multiplicity claim follows from the critical-point structure (see Section C.6 below):
 510 for $\beta \leq 1/2$, Φ is strictly increasing, so $\Phi(x) = 2\beta ac$ has at most one solution; for $\beta > 1/2$, Φ is
 511 strictly U-shaped with minimum at $x^\dagger$, so $\Phi(x) = 2\beta ac$ has at most two solutions, and the active
 512 interior root (the local minimum of R_n) is the larger. $\square$

513 C.5 Proof of Theorem 3.4 (Safe-improvement)

514 *Proof of Theorem 3.4.* The all-real benchmark uses every real observation for direct estimation,
 515 incurring risk a/n . We compare $R_n(x)$ with a/n via direct algebra. By Theorem 3.2,

$$R_n(x) = \frac{a B_n(x)}{a + (n-x) B_n(x)}.$$

516 Then $R_n(x) < a/n$ iff $a B_n(x) \cdot n < a \cdot [a + (n-x)B_n(x)]$, i.e. iff $nB_n(x) < a + (n-x)B_n(x)$,
 517 which simplifies after subtracting $(n-x)B_n(x)$ from both sides to

$$x B_n(x) < a.$$

518 The boundary case is $R_n(0) = aB_n(0)/(a + nB_n(0)) \leq a/n$ trivially (with equality only in
 519 the degenerate $B_n(0) = \infty$ limit); the restriction to $x \in (0, n)$ is what makes the iff statement
 520 non-trivial. Substituting $B_n(x) = v_n + cx^{-2\beta}$ gives the equivalent form $x(v_n + cx^{-2\beta}) < a$, i.e.
 521 $v_n x + cx^{1-2\beta} < a$. $\square$

522 C.6 Proof of Theorem 4.1 (Critical points)

523 We first establish the shape of Φ as an auxiliary lemma, then conclude about critical points of R_n .

524 **Lemma C.4** (Shape of Φ). *Under Assumption 2.3, the function $\Phi(x) = (v_n + cx^{-2\beta})^2 x^{2\beta+1}$ on*
 525 *$(0, \infty)$ satisfies:*

- 526 (a) $\Phi'(x) = x^{2\beta} B_n(x)[(2\beta+1)v_n + (1-2\beta)cx^{-2\beta}]$.
- 527 (b) *If $\beta \leq 1/2$, then $\Phi'(x) > 0$ for all $x > 0$ (with $v_n > 0$): Φ is strictly increasing, with*
 528 $\Phi(0^+) = c^2 \mathbf{1}\{\beta = 1/2\}$ *and $\Phi(\infty) = +\infty$.*
- 529 (c) *If $\beta > 1/2$, then Φ is strictly U-shaped on $(0, \infty)$ with unique minimum at*

$$x^\dagger = \left(\frac{c(2\beta-1)}{(2\beta+1)v_n} \right)^{1/(2\beta)}, \quad (18)$$

530 and minimum value

$$\Phi^\dagger := \Phi(x^\dagger) = K(\beta) c^{(2\beta+1)/(2\beta)} v_n^{(2\beta-1)/(2\beta)}, \quad K(\beta) := \frac{16\beta^2}{(2\beta+1)^{(2\beta+1)/(2\beta)} (2\beta-1)^{(2\beta-1)/(2\beta)}}. \quad (19)$$

531 $\Phi(0^+) = \Phi(\infty) = +\infty.$

532 *Proof of Theorem C.4. (a)* By the product rule applied to $\Phi(x) = B_n(x)^2 \cdot x^{2\beta+1}$,

$$\begin{aligned} \Phi'(x) &= 2B_n(x)B_n'(x)x^{2\beta+1} + B_n(x)^2(2\beta+1)x^{2\beta} \\ &= x^{2\beta} B_n(x) [2B_n'(x)x + (2\beta+1)B_n(x)]. \end{aligned}$$

533 Substituting $B_n'(x) = -2\beta cx^{-(2\beta+1)}$ and $B_n(x) = v_n + cx^{-2\beta}$,

$$\begin{aligned} 2B_n'(x)x + (2\beta+1)B_n(x) &= -4\beta cx^{-2\beta} + (2\beta+1)v_n + (2\beta+1)cx^{-2\beta} \\ &= (2\beta+1)v_n + (1-2\beta)cx^{-2\beta}. \end{aligned}$$

534 This proves (a).

535 **(b)** For $\beta \leq 1/2$ we have $1-2\beta \geq 0$ and $v_n > 0$, so the bracket $(2\beta+1)v_n + (1-2\beta)cx^{-2\beta}$ is strictly
536 positive on $(0, \infty)$. Combined with $x^{2\beta} > 0$ and $B_n(x) > 0$, this gives $\Phi'(x) > 0$ everywhere; Φ is
537 strictly increasing. The endpoint behavior follows from expanding B_n . As $x \rightarrow 0^+$, $B_n(x) \sim cx^{-2\beta}$
538 so $\Phi(x) \sim c^2 x^{1-2\beta}$; for $\beta < 1/2$ this tends to 0, while for $\beta = 1/2$, $\Phi(x) = (v_n x + c)^2 \rightarrow c^2$. As
539 $x \rightarrow \infty$, $B_n(x) \rightarrow v_n$ so $\Phi(x) \sim v_n^2 x^{2\beta+1} \rightarrow \infty$ since $v_n > 0$.

540 **(c)** For $\beta > 1/2$, $1-2\beta < 0$. The bracket $(2\beta+1)v_n + (1-2\beta)cx^{-2\beta}$ is a strictly increasing
541 function of x (since $cx^{-2\beta}$ is strictly decreasing in x and is multiplied by a negative coefficient). It
542 vanishes when $cx^{-2\beta} = (2\beta+1)v_n/(2\beta-1)$, equivalently at $x = x^\dagger$ given by Eq. (18). For $x < x^\dagger$
543 the bracket is negative, so $\Phi' < 0$; for $x > x^\dagger$ the bracket is positive, so $\Phi' > 0$. Hence Φ is strictly
544 decreasing on $(0, x^\dagger)$ and strictly increasing on $(x^\dagger, \infty)$, with unique minimum at $x^\dagger$.

545 To compute $\Phi^\dagger = \Phi(x^\dagger)$, use $cx^{\dagger-2\beta} = (2\beta+1)v_n/(2\beta-1)$, so

$$B_n(x^\dagger) = v_n + \frac{(2\beta+1)v_n}{2\beta-1} = v_n \cdot \frac{4\beta}{2\beta-1}.$$

546 From Eq. (18), $x^{\dagger 2\beta+1} = [c(2\beta-1)/((2\beta+1)v_n)]^{(2\beta+1)/(2\beta)}$. Multiplying,

$$\begin{aligned} \Phi^\dagger &= \left(\frac{4\beta v_n}{2\beta-1}\right)^2 \cdot \left(\frac{c(2\beta-1)}{(2\beta+1)v_n}\right)^{(2\beta+1)/(2\beta)} \\ &= \frac{16\beta^2 c^{(2\beta+1)/(2\beta)}}{(2\beta+1)^{(2\beta+1)/(2\beta)} (2\beta-1)^{(2\beta-1)/(2\beta)}} \cdot v_n^{(2\beta-1)/(2\beta)}, \end{aligned}$$

547 where the v_n exponent is $2 - (2\beta+1)/(2\beta) = (2\beta-1)/(2\beta)$ and the $(2\beta-1)$ exponent is
548 $(2\beta+1)/(2\beta) - 2 = (1-2\beta)/(2\beta)$, contributing the factor $(2\beta-1)^{-(2\beta-1)/(2\beta)}$ as displayed.
549 Endpoint behavior: as $x \rightarrow 0^+$, $\Phi(x) \sim c^2 x^{1-2\beta} \rightarrow +\infty$ since $1-2\beta < 0$; as $x \rightarrow \infty$,
550 $\Phi(x) \sim v_n^2 x^{2\beta+1} \rightarrow +\infty$. $\square$

551 *Proof of Theorem 4.1.* By Theorem C.3, the interior critical points of R_n on $(0, n)$ are exactly the
552 points where $\Phi(x) = 2\beta ac$, and $\text{sign}(R_n'(x)) = \text{sign}(\Phi(x) - 2\beta ac)$.

553 **(i)** $\beta \leq 1/2$. By Theorem C.4(b), Φ is strictly increasing, so the level set $\{x : \Phi(x) = 2\beta ac\}$ contains
554 at most one element. When it exists, call it x° . Then $\Phi - 2\beta ac < 0$ on $(0, x^\circ)$ and > 0 on (x°, ∞) ,
555 so $R_n' < 0$ before x° and $R_n' > 0$ after; thus x° is a strict local minimum.

556 For $\beta < 1/2$, Theorem C.4(b) gives $\Phi(0^+) = 0 < 2\beta ac < \Phi(\infty) = \infty$, so x° exists by continuity.
557 For $\beta = 1/2$, $\Phi(0^+) = c^2$ and $\Phi(\infty) = \infty$, so x° exists iff $c^2 < 2\beta ac = ac$, equivalently $c < a$.

558 **(ii)** $\beta > 1/2$. By Theorem C.4(c), Φ is U-shaped on $(0, \infty)$ with minimum value $\Phi^\dagger$ attained at $x^\dagger$.

559 *Case (a)* If $\Phi^\dagger > 2\beta ac$, then $\Phi(x) > \Phi^\dagger > 2\beta ac$ for all $x > 0$, so $R_n' > 0$ throughout $(0, n)$ and
560 there is no interior critical point.

561 *Case (b)* If $\Phi^\dagger = 2\beta ac$, $\Phi - 2\beta ac \geq 0$ with equality only at $x^\dagger$, so $R'_n \geq 0$ with $R'_n(x^\dagger) = 0$; R'_n
 562 does not change sign, so $x^\dagger$ is an inflection point but not a strict local extremum.

563 *Case (c)* If $\Phi^\dagger < 2\beta ac$: since $\Phi(0^+) = \Phi(\infty) = +\infty$ and Φ is continuous and strictly U-shaped
 564 with minimum $\Phi^\dagger < 2\beta ac$, the equation $\Phi(x) = 2\beta ac$ has exactly two solutions $x_1 \in (0, x^\dagger)$ and
 565 $x_2 \in (x^\dagger, \infty)$ by the intermediate value theorem applied to each strictly monotone branch. On $(0, x_1)$,
 566 $\Phi > 2\beta ac$ so $R'_n > 0$; on (x_1, x_2) , $\Phi < 2\beta ac$ so $R'_n < 0$; on (x_2, ∞) , $\Phi > 2\beta ac$ so $R'_n > 0$. Hence
 567 x_1 is a strict local maximum and x_2 a strict local minimum of R_n .

568 The asymptotic claim follows from Eq. (19) and Assumption 2.4: with $v_n = v_0 n^{-\rho} \{1 + o(1)\}$,

$$\Phi^\dagger = K(\beta) c^{(2\beta+1)/(2\beta)} v_0^{(2\beta-1)/(2\beta)} n^{-\rho(2\beta-1)/(2\beta)} \{1 + o(1)\},$$

569 which tends to 0 when $\rho > 0$ and $\beta > 1/2$, since the exponent $-\rho(2\beta-1)/(2\beta)$ is strictly negative.
 570 Hence $\Phi^\dagger < 2\beta ac$ for all sufficiently large n , putting us in case (iic). $\square$

571 C.7 Proof of Theorem 4.2 (Phase diagram)

572 *Proof of Theorem 4.2.* We analyze the polynomial form of the FOC,

$$v_n^2 x^{2\beta+1} + 2cv_n x + c^2 x^{1-2\beta} = 2\beta ac, \quad (20)$$

573 in each regime, using Theorem 4.1 for existence and identification of the active root, and the global
 574 minimization $x_n^* \in \arg \min_{x \in [0, n]} R_n(x)$.

575 **Regime I** ($\rho = 0$, $v_n \rightarrow v_0 > 0$). With $v_n \rightarrow v_0$, the FOC Eq. (20) reads $v_0^2 x^{2\beta+1} + 2cv_0 x +$
 576 $c^2 x^{1-2\beta} = 2\beta ac + o(1)$, which has no n -dependence in the leading coefficients. By Theorem 4.1,
 577 any active interior root is determined by the constants (a, c, v_0, β) alone and is therefore $O(1)$. Since
 578 the budget n grows unboundedly, $x_n^\circ/n \rightarrow 0$. The candidates $\{0, x_n^\circ, n\}$ are evaluated by direct
 579 substitution into R_n ; the global minimizer is one of $\{0, x_n^\circ\}$ since $R_n(n) = v_n + cn^{-2\beta} \rightarrow v_0$ does
 580 not vanish, while $R_n(0) = a/n \rightarrow 0$. Hence $x_n^* \in \{0, x_n^\circ\}$ and in either case $\lambda_n^* = x_n^*/n \rightarrow 0$.

581 **Regime II** ($\beta < 1/2$, $\rho > 0$). Now $v_n \rightarrow 0$. We claim $x_n^\circ = \Theta(1)$. With x bounded, $v_n^2 x^{2\beta+1} =$
 582 $O(v_n^2)$ and $2cv_n x = O(v_n)$, both $o(1)$, so Eq. (20) reduces to $c^2 x^{1-2\beta} = 2\beta ac + o(1)$. Solving,
 583 $x^{1-2\beta} = 2\beta a/c + o(1)$, so $x_n^\circ \rightarrow x_\infty := (2\beta a/c)^{1/(1-2\beta)}$. To rule out the alternative $x_n^\circ \rightarrow \infty$,
 584 observe that if so, then because $1 - 2\beta > 0$, the term $c^2 x^{1-2\beta} \rightarrow \infty$, contradicting Eq. (20) (whose
 585 right side is finite). So indeed $x_n^\circ = \Theta(1)$, hence $x_n^\circ/n \rightarrow 0$.

586 For the rate: $V_R(x_n^\circ) = a/(n - x_n^\circ) \sim a/n$ since $x_n^\circ = O(1) \ll n$. And $B_n(x_n^\circ) = v_n + cx_n^{\circ-2\beta} \rightarrow$
 587 $cx_\infty^{-2\beta} > 0$ (the v_n term vanishes). Substituting into the harmonic-mean form of R_n ,

$$R_n(x_n^\circ) = \frac{V_R B_n}{V_R + B_n} = V_R \cdot \frac{1}{1 + V_R/B_n} = V_R - \frac{V_R^2}{B_n} + O((V_R/B_n)^3),$$

588 which yields $R_n(x_n^\circ) = a/n - (a/n)^2/(cx_\infty^{-2\beta}) + o(n^{-2})$ as displayed in the theorem. Safe-
 589 improvement:

$$x_\infty B_n(x_\infty) \rightarrow x_\infty \cdot cx_\infty^{-2\beta} = cx_\infty^{1-2\beta} = c \cdot (2\beta a/c) = 2\beta a < a$$

590 since $\beta < 1/2$. Hence by Theorem 3.4, $R_n(x_n^\circ) < a/n$ for n large, and the global oracle is
 591 $x_n^* = x_n^\circ \{1 + o(1)\}$.

592 **Regime III** ($\beta > 1/2$, $0 < \rho < \beta + 1/2$). By Theorem 4.1(iic), the active interior root x_n° exists for all
 593 n sufficiently large. We make and verify the conjecture that the bias term is asymptotically negligible
 594 relative to the synthetic-variance term at the root: $cx_n^{\circ-2\beta} \ll v_n$, hence $B_n(x_n^\circ) = v_n \{1 + o(1)\}$.
 595 Substituting this into the FOC Eq. (4) gives

$$v_n^2 = 2\beta ac x_n^{\circ-(2\beta+1)} \{1 + o(1)\},$$

596 which solves to

$$\begin{aligned} x_n^\circ &= \left(\frac{2\beta ac}{v_n^2} \right)^{1/(2\beta+1)} \{1 + o(1)\} \\ &= \left(\frac{2\beta ac}{v_0^2} \right)^{1/(2\beta+1)} n^{2\rho/(2\beta+1)} \{1 + o(1)\}, \end{aligned}$$

597 using $v_n = v_0 n^{-\rho} \{1 + o(1)\}$ to substitute. This is the displayed expression Eq. (9).

598 To verify the conjecture, compute the ratio of the two terms in B_n :

$$\begin{aligned} \frac{cx_n^{\circ-2\beta}}{v_n} &= c \cdot \left[\frac{2\beta ac}{v_0^2} \right]^{-2\beta/(2\beta+1)} n^{-4\beta\rho/(2\beta+1)} \cdot \frac{1}{v_0 n^{-\rho}} \{1 + o(1)\} \\ &\asymp n^{-4\beta\rho/(2\beta+1)+\rho} = n^{\rho \cdot [1-4\beta/(2\beta+1)]} = n^{-\rho(2\beta-1)/(2\beta+1)}, \end{aligned}$$

599 which tends to 0 as $n \rightarrow \infty$ since $\beta > 1/2$ and $\rho > 0$. Hence $cx_n^{\circ-2\beta} = o(v_n)$, confirming the
600 conjecture.

601 Feasibility: $x_n^{\circ}/n \asymp n^{2\rho/(2\beta+1)-1}$. The exponent $2\rho/(2\beta+1) - 1$ is strictly negative iff $\rho < \beta + 1/2$,
602 which is the assumed range. Hence $x_n^{\circ} = o(n)$, so $\lambda_n^{\circ} \rightarrow 0$.

603 Safe-improvement: $x_n^{\circ} B_n(x_n^{\circ}) \asymp n^{2\rho/(2\beta+1)} \cdot n^{-\rho} = n^{-\rho(2\beta-1)/(2\beta+1)} \rightarrow 0 < a$, so by Theo-
604 rem 3.4, $R_n(x_n^{\circ}) < a/n$ for n large. We also have $R_n(x_n^{\circ}) < R_n(n)$ in this regime by Regime IV's
605 analysis below (strict inequality at the boundary). Hence $x_n^{\star} = x_n^{\circ} \{1 + o(1)\}$.

606 For the risk asymptotics: $V_R(x_n^{\circ}) = a/(n - x_n^{\circ}) = a/n \{1 + o(1)\}$ since $x_n^{\circ} = o(n)$, and $B_n(x_n^{\circ}) =$
607 $v_n \{1 + o(1)\}$. The harmonic-mean formula

$$R_n(x_n^{\star}) = \frac{V_R B_n}{V_R + B_n} \{1 + o(1)\} = \frac{(a/n) v_n}{a/n + v_n} \{1 + o(1)\}$$

608 yields the three subcases by direct substitution: if $\rho < 1$, $v_n \gg a/n$ so the denominator is dominated
609 by v_n and $R_n \sim a/n$; if $\rho = 1$, both terms are of the same order $1/n$ and $R_n \sim av_0/[(a + v_0)n]$; if
610 $\rho > 1$, $v_n \ll a/n$ so the denominator is dominated by a/n and $R_n \sim v_n = v_0/n^{\rho}$.

611 **Regime IV** ($\beta > 1/2, \rho > \beta + 1/2$). The Regime-III asymptotic formula gives $x_n^{\circ} \asymp n^{2\rho/(2\beta+1)}$.
612 Here $2\rho/(2\beta+1) > 1$, so $x_n^{\circ} > n$ for n large and the interior root is infeasible. We compare
613 the boundary candidates $R_n(0) = a/n$ and $R_n(n) = v_n + cn^{-2\beta}$. With $\rho > \beta + 1/2 > 1$ and
614 $2\beta > 1$, $v_n = O(n^{-\rho}) = o(n^{-1})$ and $cn^{-2\beta} = o(n^{-1})$, so $R_n(n) = o(n^{-1}) < a/n$ eventually.
615 Hence the global oracle is at the right boundary: $x_n^{\star} = n \{1 + o(1)\}$, $\lambda_n^{\star} \rightarrow 1$, and the risk rate
616 $R_n(x_n^{\star}) = v_n + cn^{-2\beta} = o(n^{-1})$ beats the all-real rate.

617 **Regime V** ($\beta = 1/2$). With $\beta = 1/2$, the FOC Eq. (4) becomes $(v_n + cx^{-1})^2 = acx^{-2}$, i.e.,
618 $(v_n x + c)^2 = ac$ after multiplying through by x^2 . Taking the positive square root (the negative root
619 $v_n x + c = -\sqrt{ac}$ gives $x < 0$, infeasible), $v_n x + c = \sqrt{ac}$, so

$$x_n^{\circ} = \frac{\sqrt{ac} - c}{v_n},$$

620 which is positive iff $\sqrt{ac} > c$, equivalently $a > c$. The asymptotic order follows from $v_n =$
621 $v_0 n^{-\rho} \{1 + o(1)\}$, giving $x_n^{\circ} = ((\sqrt{ac} - c)/v_0) n^{\rho} \{1 + o(1)\}$.

622 The boundary regime depends on ρ :

- 623 • If $\rho < 1$, $x_n^{\circ} = o(n)$, so $\lambda_n^{\circ} \rightarrow 0$ and $R_n(x_n^{\star}) \sim a/n$ with strict finite-sample improvement (the
624 shrinkage gain $-(a/n)^2/B_n(x_n^{\circ})$ is strictly negative).
- 625 • If $\rho = 1$, $\lambda_n^{\circ} \rightarrow (\sqrt{ac} - c)/v_0 \in (0, 1)$ provided $\sqrt{ac} - c < v_0$, with $R_n(x_n^{\star}) \asymp n^{-1}$.
- 626 • If $\rho > 1$, x_n° exceeds n asymptotically; we compare boundaries. $R_n(n) = v_n + c/n \sim c/n$ since
627 $\rho > 1$ makes $v_n = o(1/n)$. This beats $R_n(0) = a/n$ iff $c < a$, which holds by hypothesis. Hence
628 $\lambda_n^{\star} \rightarrow 1$ and $R_n(x_n^{\star}) = v_n + c/n = O(\max(n^{-\rho}, n^{-1}))$.

629 If instead $a \leq c$, then $\sqrt{ac} \leq c$, so the formula gives $x_n^{\circ} \leq 0$, so there is no interior root. By
630 Theorem 3.4, the safe-improvement condition at $x > 0$ becomes $v_n x + c < a$; since $c \geq a$ this fails
631 for any $x \geq 0$, so the boundary $x_n^{\star} = 0$ dominates. This concludes Regime V and the proof. $\square$

632 C.8 Proof of Theorem A.1 (Plateau bias)

633 *Proof of Theorem A.1.* Replace Assumption 2.3 by $b_2(x) = b_\infty^2 + cx^{-2\beta}\{1 + o(1)\}$ with $b_\infty^2 > 0$
 634 and $\beta > 0$, and keep Assumption 2.4 with $\rho \geq 0$. Now $B_n(x) = v_n + b_\infty^2 + cx^{-2\beta} \geq b_\infty^2 > 0$ for
 635 all $x > 0$, so B_n is bounded below by a strictly positive constant uniformly in x and n .

636 The FOC $B_n(x)^2 = 2\beta ac x^{-(2\beta+1)}$ from Theorem 3.3 implies in particular

$$b_\infty^4 \leq B_n(x)^2 = 2\beta ac x^{-(2\beta+1)},$$

637 which yields the explicit upper bound $x^{2\beta+1} \leq 2\beta ac/b_\infty^4$, equivalently

$$x_n^\circ \leq \left(\frac{2\beta ac}{b_\infty^4} \right)^{1/(2\beta+1)} = O(1).$$

638 This is Eq. (12), and it shows $x_n^\circ = O(1)$ regardless of β, ρ . Consequently $\lambda_n^\circ = x_n^\circ/n \rightarrow 0$ for any
 639 $\beta > 0$ and $\rho \geq 0$.

640 For the risk: $V_R(x_n^\circ) = a/(n - x_n^\circ) \sim a/n$, since x_n° is bounded. And $B_n(x_n^\circ) \geq b_\infty^2 > 0$ is
 641 bounded away from zero. Hence $V_R \ll B_n$ asymptotically. As in Regime II,

$$R_n(x_n^\circ) = V_R - V_R^2/B_n + o(V_R^2/B_n) = \frac{a}{n} - \frac{(a/n)^2}{B_n(x_n^\circ)} + o(n^{-2}),$$

642 so $R_n(x_n^\circ) = a/n\{1 + o(1)\}$, giving only finite-sample shrinkage gain, no first-order rate im-
 643 provement. The shrinkage is strictly positive iff $R_n(x_n^\circ) < a/n$, equivalent by Theorem 3.4 to
 644 $x_n^\circ B_n(x_n^\circ) < a$, which becomes $x_n^\circ b_\infty^2 < a$ at leading order (since v_n and $cx_n^{-2\beta}$ are $o(1)$ relative
 645 to b_∞^2 when $b_\infty^2 > 0$ and x_n° is bounded). $\square$

646 C.9 Proof of Theorem 5.1 (Adaptive oracle inequality)

647 *Proof of Theorem 5.1.* Let $\mathcal{G}_n^+ = \mathcal{G}_n \cup \{0, n\}$ and recall the oracle and adaptive choices

$$x_n^\star := \arg \min_{x \in \mathcal{G}_n^+} R_n(x), \quad \hat{x}_n := \arg \min_{x \in \mathcal{G}_n^+} \hat{R}_n(x),$$

648 ties broken arbitrarily. Fix $\delta > 0$ arbitrary. On the event $\{\Delta_n \leq \delta\}$, the uniform-relative-consistency
 649 bound $\sup_{x \in \mathcal{G}_n^+} |\hat{R}_n(x)/R_n(x) - 1| \leq \delta$ implies the two-sided sandwich

$$(1 - \delta) R_n(x) \leq \hat{R}_n(x) \leq (1 + \delta) R_n(x) \quad \text{for every } x \in \mathcal{G}_n^+. \quad (21)$$

650 By definition of $\hat{x}_n$ as the empirical minimizer, $\hat{R}_n(\hat{x}_n) \leq \hat{R}_n(x_n^\star)$. Chaining Eq. (21) applied at
 651 $x = \hat{x}_n$ on the left and $x = x_n^\star$ on the right with this minimization,

$$(1 - \delta) R_n(\hat{x}_n) \leq \hat{R}_n(\hat{x}_n) \leq \hat{R}_n(x_n^\star) \leq (1 + \delta) R_n(x_n^\star).$$

652 Dividing by $(1 - \delta)R_n(x_n^\star)$ (both factors are strictly positive),

$$\frac{R_n(\hat{x}_n)}{R_n(x_n^\star)} \leq \frac{1 + \delta}{1 - \delta}.$$

653 Conversely, $R_n(\hat{x}_n)/R_n(x_n^\star) \geq 1$ trivially since $x_n^\star$ is the oracle minimizer over the same grid.
 654 Hence on $\{\Delta_n \leq \delta\}$,

$$1 \leq \frac{R_n(\hat{x}_n)}{R_n(x_n^\star)} \leq \frac{1 + \delta}{1 - \delta}.$$

655 Since $\Delta_n \xrightarrow{p} 0$ by hypothesis, for any fixed $\eta > 0$ we can choose δ small so that $(1 + \delta)/(1 - \delta) < 1 + \eta$
 656 and conclude

$$\mathbb{P}\left(|R_n(\hat{x}_n)/R_n(x_n^\star) - 1| > \eta\right) \leq \mathbb{P}(\Delta_n > \delta) \rightarrow 0,$$

657 which is the conclusion of Theorem 5.1. The continuous-oracle claim follows by combining
 658 $R_n(x_n^\star) = (1 + o(1)) \inf_{[0, n]} R_n$ (the grid near-optimality hypothesis) with the previous display and
 659 Slutsky. $\square$

660 C.10 Proof of Proposition B.1 (Centered CLT)

661 *Proof of Proposition B.1.* By Lemma C.1(i) and Assumption 2.1 and 2.2, conditionally on $\mathcal{F}_C$ (the
662 calibration σ -field) the linear estimator decomposes as

$$\hat{\theta}^L(\alpha^*, x) - \theta^* = \alpha^*(\hat{\theta}_R - \theta^*) + (1 - \alpha^*)(\hat{\theta}_S - \theta_x) + (1 - \alpha^*)\Delta(x),$$

663 where $\Delta(x) := \theta_x - \theta^*$ is the calibration-induced bias, which is $\mathcal{F}_C$ -measurable. The first two
664 summands are independent zero-mean martingale-difference averages with conditional variances
665 $\alpha^{*2}V_R(x)\{1 + o(1)\}$ and $(1 - \alpha^*)^2v_n\{1 + o(1)\}$ respectively (using Assumption 2.1 and 2.2 and
666 the synthetic-error decomposition). They are conditionally independent because $\hat{\theta}_R$ depends only on
667 E_x and $\hat{\theta}_S$ depends only on the synthetic draws given $\mathcal{F}_C$ (with E_x and the synthetic draws mutually
668 independent).

669 Define $\Omega_n(x) := \alpha^{*2}V_R(x) + (1 - \alpha^*)^2v_n$ as the stochastic-variance contribution. Subtracting the
670 bias from both sides and dividing by $\sqrt{\Omega_n(x_n^*)}$,

$$\frac{\hat{\theta}^L - \theta^* - (1 - \alpha^*)\Delta(x_n^*)}{\sqrt{\Omega_n(x_n^*)}} = \frac{\alpha^*(\hat{\theta}_R - \theta^*) + (1 - \alpha^*)(\hat{\theta}_S - \theta_{x_n^*})}{\sqrt{\Omega_n(x_n^*)}}.$$

671 The right side is a sum of two independent triangular arrays with combined conditional variance
672 equal to 1. The Lindeberg-Feller CLT applied conditionally on $\mathcal{F}_C$ gives convergence to $N(0, 1)$
673 provided the Lindeberg condition holds for ψ and ϕ_x (the influence functions of $\hat{\theta}_R$ and $\hat{\theta}_S$), which is
674 assumed. Convergence holds unconditionally as well since the limiting distribution is non-random,
675 allowing exchange of conditional and unconditional limits via the dominated convergence theorem
676 applied to characteristic functions.

677 Replacing the oracle weights and oracle allocation by their adaptive estimates $(\hat{\alpha}, \hat{x}_n)$ from Theo-
678 rem 5.1 preserves the CLT under the same regularity: by Theorem 5.1 and the continuity of $\alpha^*(\cdot)$,
679 $\hat{\alpha} = \alpha^*(x_n^*)\{1 + o_p(1)\}$ and $\Omega_n(\hat{x}_n)/\Omega_n(x_n^*) \xrightarrow{P} 1$, so Slutsky's theorem yields the same Gaussian
680 limit. $\square$

681 C.11 Proof of Corollary B.2 (Wald inference)

682 *Proof of Corollary B.2.* A two-sided Wald interval of width $z_{1-\alpha/2}\sqrt{\Omega_n(x_n^*)}$ centered at $\hat{\theta}^L$ has
683 asymptotic coverage $1 - \alpha$ for θ^* if and only if the standardized residual $(\hat{\theta}^L - \theta^*)/\sqrt{\Omega_n(x_n^*)}$
684 converges to $N(0, 1)$. By Proposition B.1, the standardized residual centered at $\theta^* + (1 - \alpha^*)\Delta(x_n^*)$
685 converges to $N(0, 1)$. Removing the centering shift gives

$$\frac{\hat{\theta}^L - \theta^*}{\sqrt{\Omega_n(x_n^*)}} = \frac{\hat{\theta}^L - \theta^* - (1 - \alpha^*)\Delta(x_n^*)}{\sqrt{\Omega_n(x_n^*)}} + \frac{(1 - \alpha^*)\Delta(x_n^*)}{\sqrt{\Omega_n(x_n^*)}}.$$

686 The first summand converges to $N(0, 1)$. The second is a deterministic shift in the limit. The
687 standardized residual is asymptotically standard normal, and the Wald interval has correct asymptotic
688 coverage, if and only if this second term tends to zero in probability:

$$(1 - \alpha^*(x_n^*))\Delta(x_n^*) = o_p(\sqrt{\Omega_n(x_n^*)}). \quad (22)$$

689 This proves the iff statement. Now examine Eq. (22) in two specific regimes.

690 *Regime III with $\rho = 1$, $\beta > 1/2$.* By Theorem 4.2 (Regime III with $\rho = 1$), $x_n^* \asymp n^{2/(2\beta+1)}$ and
691 $R_n(x_n^*) \asymp n^{-1}$, so $\Omega_n(x_n^*) \asymp n^{-1}$. The calibration bias satisfies $\mathbb{E}[\Delta(x_n^*)^2] = b^2(x_n^*) \asymp x_n^{*-2\beta} \asymp$
692 $n^{-4\beta/(2\beta+1)}$. Since $4\beta/(2\beta+1) > 1$ iff $\beta > 1/2$, we have $\Delta(x_n^*) = O_p(n^{-2\beta/(2\beta+1)}) =$
693 $o(n^{-1/2}) = o(\sqrt{\Omega_n(x_n^*)})$. Multiplying by $1 - \alpha^* = O(1)$ preserves the order, so Eq. (22) holds.

694 *Regime V with $\rho = 1$, $a > c$, $\beta = 1/2$.* In this regime $x_n^* \asymp n$ but $\lambda_n^* \rightarrow (\sqrt{ac} - c)/v_0 \in (0, 1)$, so
695 $\Omega_n(x_n^*) \asymp n^{-1}$. The bias satisfies $b^2(x_n^*) \asymp x_n^{*-1} \asymp n^{-1}$, so $\Delta(x_n^*)$ is the same order as $\sqrt{\Omega_n(x_n^*)}$.
696 Generically $1 - \alpha^*$ is bounded away from zero, so Eq. (22) fails: $(1 - \alpha^*)\Delta(x_n^*)$ is $O_p(\sqrt{\Omega_n(x_n^*)})$,
697 not o_p . Hence the Wald interval suffers a non-vanishing standardized bias and undercovers in the
698 limit. $\square$

Table 1: Theoretical vs. empirical allocation scaling slopes by (β, ρ) .

β	ρ	theory slope	empirical slope	SE
0.75	0.5	0.400	0.453	0.006
0.75	1.0	0.800	0.842	0.003
1.00	0.5	0.333	0.375	0.011
1.00	1.0	0.667	0.684	0.003
1.50	0.5	0.250	0.250	0.006
1.50	1.0	0.500	0.504	0.002
2.00	0.5	0.200	0.204	0.014
2.00	1.0	0.400	0.404	0.003

^a Experiment 1 full profile; slopes estimated by OLS of $\log x_n^*$ on $\log n$ across $n \in \{100, \dots, 20,000\}$.

D Adaptive estimator pseudocode

The algorithm box below was moved from Section 5.1 (formerly displayed inline in the main paper) to fit the 9-page main-content limit; it is reproduced here verbatim.

Algorithm 1 Corrected adaptive Generalized-Neyman estimator.

- 1: **Input:** real sample $X_1, \dots, X_n$; calibratable generator $\mathcal{G}$; pilot grid $\mathcal{G}_{\text{pilot}}$; full grid $\mathcal{G}_n^+ := \mathcal{G}_n \cup \{0, n\}$; number of folds K .
 - 2: Cross-fit split $[n]$ into K folds; for each fold, allocate out-of-fold data into calibration-pilot, validation, and final-estimation slices. Loop over folds.
 - 3: Estimate $\hat{a}$ as the empirical variance of the influence-function values $\psi(X_i)$ on the out-of-fold slice.
 - 4: **for** each $x_j \in \mathcal{G}_{\text{pilot}}$ **do**
 - 5: Calibrate $\mathcal{G}$ on x_j pilot observations, draw m_n synthetic points, compute $\hat{\theta}_S(x_j)$, and estimate $\hat{v}_n$ as the synthetic sample variance divided by m_n .
 - 6: Validation-debias the squared bias: $\hat{b}^2(x_j) = [(\hat{\theta}_S(x_j) - \hat{\theta}_R^V)^2 - \widehat{\text{Var}}(\hat{\theta}_S(x_j)) - \widehat{\text{Var}}(\hat{\theta}_R^V)]_+$, where $\hat{\theta}_R^V$ is the real estimator on the validation slice.
 - 7: **end for**
 - 8: Fit a power law by ordinary least squares: $\log \hat{b}^2(x_j) = \log \hat{c} - 2\hat{\beta} \log x_j + \epsilon_j$.
 - 9: Form $\hat{B}_n(x) = \hat{v}_n + \hat{c} x^{-2\hat{\beta}}$, $\hat{V}_R(x) = \hat{a}/(n-x)$, and $\hat{R}_n(x) = \hat{V}_R(x)\hat{B}_n(x)/[\hat{V}_R(x) + \hat{B}_n(x)]$ on $\mathcal{G}_n^+$, with $\hat{R}_n(0) = \hat{a}/n$ and $\hat{R}_n(n) = \hat{B}_n(n)$.
 - 10: $\hat{x}_n \leftarrow \arg \min_{x \in \mathcal{G}_n^+} \hat{R}_n(x)$.
 - 11: **Safety check:** if $\hat{x}_n \hat{B}_n(\hat{x}_n) \geq \hat{a}$, return the all-real estimator $\hat{\theta}_R$.
 - 12: Otherwise set $\hat{\alpha} = \hat{B}_n(\hat{x}_n)/[\hat{V}_R(\hat{x}_n) + \hat{B}_n(\hat{x}_n)]$ and return $\hat{\theta} = \hat{\alpha} \hat{\theta}_R(\hat{x}_n) + (1 - \hat{\alpha}) \hat{\theta}_S(\hat{x}_n)$.
 - 13: **Output:** cross-fit average of $\hat{\theta}$ over the K folds.
-

E Additional experiments

This appendix collects three supporting experiments referenced in Section 6. All three are run at the full parameter sweep (Experiment 3 with 48,000 raw rows, Experiment A with 72,000, and Experiment B with 14,400). These experiments test whether the corrected allocation logic extends beyond the scalar Gaussian DGP and show where the adaptive estimator’s safety fallback engages.

E.1 Complete summaries for main experiments

Experiment 1 scaling details. Table 1 reports the representative scaling slopes behind Section 6.1. The four upper rows are the cells where empirical slopes exceeded theory; the four lower rows are high- β cells where the empirical and theoretical slopes agree closely.

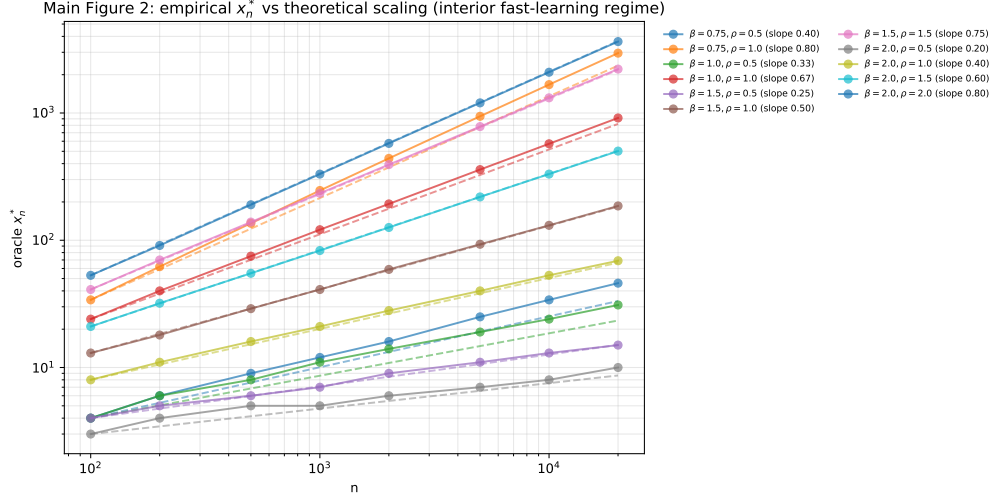


Figure 2: Empirical and theoretical scaling of the oracle calibration size $\log x_n^*$ versus $\log n$ for four representative regimes (fixed m ; slow learner $\beta = 0.4, \rho = 1$; common fast learner $\beta = 1, \rho = 1$; boundary $\beta = 1, \rho = 2$). Solid lines are theoretical slopes from Theorem 4.2; markers are oracle solutions to the corrected first-order condition on the simulation grid.

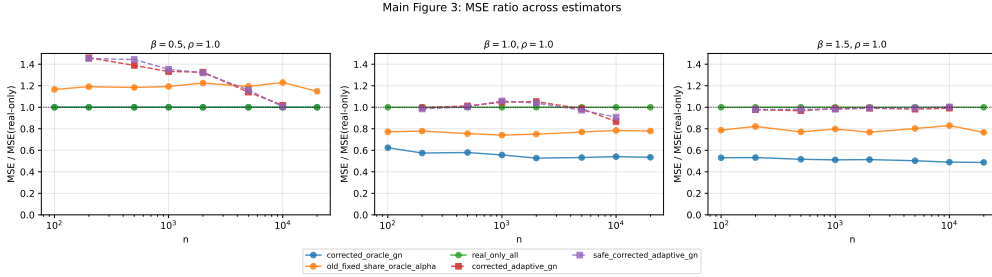


Figure 3: MSE ratio to `real_only_all` across n for the corrected oracle, the safe corrected oracle (with the safety check from Theorem 3.4), naive pooling, the equal-split baseline, the rejected old fixed-share rule, and synthetic-only with full calibration. The corrected oracle is the lower envelope across the entire grid; the old fixed-share rule pays a constant-factor penalty in the fast-learning regime by over-calibrating.

Table 2: Adaptive estimator MSE ratios at the headline cell ($\beta = 1, \rho = 1, n = 10,000$) and aggregate harm statistics across the 60 non-trivial cells of the Experiment 2 grid.

method	headline MSE	headline regret	harm cells	max ratio
corrected_oracle_gn	0.522	0.000	—	—
corrected_adaptive_gn	0.867	0.663	21/60	1.460
safe_corrected_adaptive_gn	0.908	0.741	22/60	1.453
adaptive_parametric_foc	0.874	0.675	29/60	1.432
adaptive_nonparametric_grid	1.136	1.177	44/60	1.834

^a “Harm” means MSE ratio to `real_only_all` exceeds one. Oracle regret is $\text{MSE}/\text{MSE}(\text{corrected_oracle_gn}) - 1$.

711 **Experiment 2 adaptive details.** Table 2 gives the headline adaptive MSE ratios and aggregate
712 harm counts. The table is meant to show finite-sample limitations, not only successes: the adaptive
713 estimator is stable relative to naive synthetic-use baselines, but it is not oracle-equivalent on this grid.

714 **Experiment 4 coverage details.** Table 3 reports the full-grid coverage range for each interval
715 method.

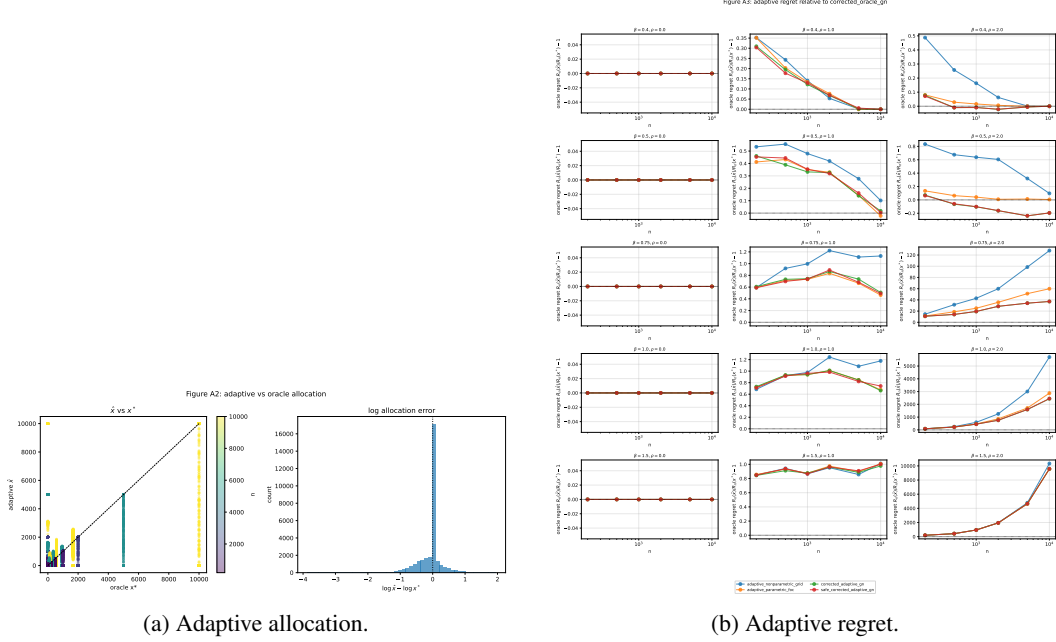


Figure 4: *Left*: adaptive allocation $\hat{x}_n$ against oracle x_n° across (β, ρ, n) cells. Points along the diagonal indicate close allocation tracking. *Right*: adaptive regret $R_n(\hat{x}_n)/R_n(x_n^\circ) - 1$ as a function of n . The corrected adaptive estimator with safety fallback (Theorem 5.1) usually selects a similar calibration size but still has a visible finite-sample MSE gap at $n \leq 10^4$.

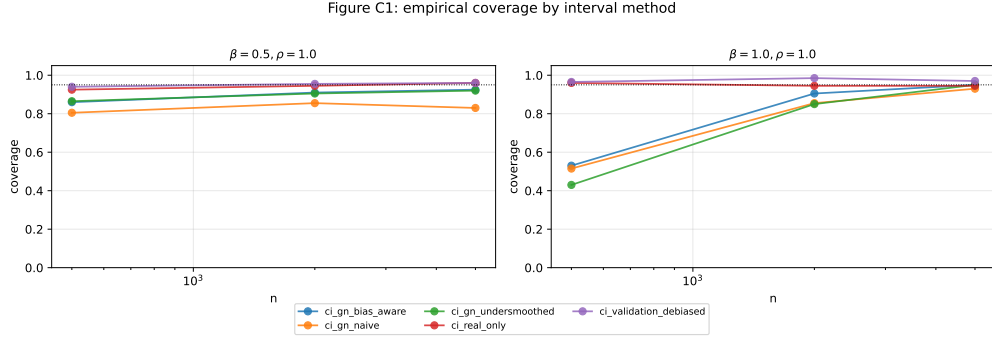


Figure 5: Empirical coverage of 95% intervals against n for five interval methods at $\rho = 1$ across $\beta \in \{0.4, 0.5, 0.75, 1.0\}$. Dashed line at nominal 0.95. Naive Wald around the MSE-optimal estimator (`ci_gn_naive`) undercovers when synthetic bias is non-negligible (naive coverage min 0.855 vs validation-debiased min 0.938 across the full Experiment 4 grid). Validation debiasing (`ci_validation_debiased`) and the all-real interval (`ci_real_only`) sit closest to nominal coverage; bias-aware widening and undersmoothing only partially close the gap.

E.2 Multichannel allocation (Experiment 3)

Setup. Experiment 3 generalises the working model to two calibration channels with effective synthetic error

$$B_n(x_1, x_2) = v_n + c_1 x_1^{-2\beta_1} + c_2 x_2^{-2\beta_2}, \quad V_R(x_1, x_2) = \frac{a}{n - x_1 - x_2},$$

so each channel has its own learning exponent and bias-decay constant. Defaults are $a = c_1 = c_2 = \sigma_S^2 = 1$ and $m_n = n$ (so $\rho = 1$), and the learning-exponent pairs are $(\beta_1, \beta_2) \in \{(0.5, 1.0), (0.75, 1.5), (0.4, 1.0)\}$. Channel 1 plays the role of a slow channel (e.g. retrieval conditioning) and channel 2 the role of a fast channel (e.g. fine-tuning). The full Experiment 3 sweep produces 48,000 raw replication rows.

Table 3: Coverage by interval method across the full Experiment 4 grid.

interval method	coverage min	coverage median	coverage max
ci_gn_naive	0.855	0.916	0.955
ci_gn_bias_aware	0.896	0.930	0.957
ci_gn_undersmoothed	0.913	0.936	0.955
ci_real_only	0.938	0.949	0.956
ci_validation_debiased	0.938	0.953	0.964

^a Experiment 4 full profile (280,000 replications), nominal level 95%. Min/median/max are taken across all (β, ρ, n) cells in the grid.

Table 4: Multichannel performance summary (Experiment 3, full profile). Median and maximum MSE ratio to `corrected_multichannel_oracle` across the (β_1, β_2, n) grid.

method	median MSE ratio	max MSE ratio
<code>corrected_multichannel_oracle</code>	1.000	1.000
<code>best_single_channel_oracle</code>	1.147	1.666
<code>equal_split_two_channels</code>	1.493	1.745
<code>old_multichannel_fixed_share</code>	2.240	3.217

^a Full profile (48,000 raw rows) across three (β_1, β_2) pairs and the n grid; `corrected_multichannel_oracle` is the per-cell benchmark, ratios are aggregated across cells. No cell beats the corrected multichannel oracle in Monte Carlo MSE.

KKT verification. The corrected multichannel oracle equalises the per-channel marginal value at any interior optimum: for every active channel k ,

$$MV_k(\mathbf{x}) = 2\beta_k a c_k x_k^{-(2\beta_k+1)} = B_n(\mathbf{x})^2,$$

which is the natural KKT generalisation of the single-channel FOC in Theorem 3.3. Intuitively, at an interior optimum the marginal MSE reduction from spending one more real observation on calibration must be the same in every active channel; otherwise reallocating one observation from a low- MV channel to a high- MV channel would strictly improve risk. The corrected multichannel oracle realises this equality by construction, while the old multichannel fixed-share rule $\lambda_k^{\text{old}} = 2\beta_k / (1 + 2 \sum_j \beta_j)$ generally does not. Figure 6a plots the two-channel risk surface $R_n(x_1, x_2)$ with each method’s selected allocation overlaid, and Figure 6b plots MV_1 versus MV_2 at the selected allocations.

Results. Table 4 reports the median and maximum MSE ratio of each method to the corrected multichannel oracle across the full grid. The corrected oracle dominates: `best_single_channel_oracle` sits at median 1.147, `equal_split_two_channels` at 1.493, and `old_multichannel_fixed_share` at 2.240. The fixed-share rule’s roughly $2\times$ MSE inflation matches the qualitative prediction of Theorem 4.2: when both channels learn fast and $\rho = 1$, the optimal share in each channel vanishes with n , while the fixed-share rule allocates a constant fraction of real data to calibration regardless of n . No full-profile cell beats the corrected multichannel oracle in Monte Carlo MSE.

E.3 Tabular semi-synthetic benchmark (Experiment A)

Setup. Experiment A tests the adaptive method on the UCI Adult / Census Income tabular dataset using the pseudo-population protocol of Section 6: the cleaned full dataset is treated as $\mathcal{D}_{\text{pop}}$, real samples of size n are drawn without replacement, and the ground truth is the pseudo-population value of the estimand. We report two estimands: the binary income outcome mean $\theta_1^* = \mathbb{E}[Y]$ and the subgroup gap $\theta_2^* = \mathbb{E}[Y \mid G = 1] - \mathbb{E}[Y \mid G = 0]$. The synthetic generator is the Gaussian-copula baseline (`gaussian_copula`, with eigenvalue-clipped correlation matrix to guard against rank-deficient calibration samples). The grid is $n \in \{200, 500, 1000, 2000, 5000\}$ and $m \in \{n, 5n, 10n\}$. The full Experiment A sweep produces 72,000 raw rows across 45 retained cells.

Adaptive behaviour. Across the 45 retained cells of the tabular grid, both `corrected_adaptive_gn` and `safe_corrected_adaptive_gn` fall back to the all-real es-

Table 5: Tabular semi-synthetic performance summary (Experiment A, full profile). Aggregates across the 45 retained cells of the $n \times m$ grid for the Adult-pseudo-population mean and subgroup gap.

method	cells	median MSE ratio	max MSE ratio	harmful cells	mean fallback
corrected_adaptive_gn	45	1.000	1.000	0/45	1.000
safe_corrected_adaptive_gn	45	1.000	1.000	0/45	1.000
validation_debiased_gn	45	1.000	1.000	0/45	1.000
old_fixed_share_plugin_alpha	36	1.294	1.501	36/36	0.000
fixed_half_split_plugin_alpha	45	3.519	4.316	45/45	0.000
naive_pooling	45	5.439	117.166	45/45	0.000
synthetic_only_full_calibration	45	8.874	141.827	45/45	0.000

^a Full profile (72,000 raw rows); UCI Adult dataset; gaussian_copula generator with eigenvalue-clipped correlation matrix. “Harmful” means MSE ratio to `real_only_all` strictly exceeds one. The adaptive variants engage the safe fallback in all 45 cells, so their MSE matches `real_only_all` exactly. The `old_fixed_share_plugin_alpha` row reports 36 cells because the plug-in $\hat{\alpha}$ is undefined in the 9 cells where the plug-in bias estimate is non-positive.

timator in 45/45 cells, with median and maximum MSE ratio to `real_only_all` of exactly 1.000 and zero harmful cells (mean fallback rate 1.000). `validation_debiased_gn` also returns the all-real point estimate in 45/45 cells. In contrast, the rejected and naive baselines are uniformly harmful: `old_fixed_share_plugin_alpha` sits at median 1.294 (max 1.501, harmful in 36/36 evaluable cells), `fixed_half_split_plugin_alpha` at median 3.519 (max 4.316, harmful in 45/45), `naive_pooling` at median 5.439 (max 117.166, harmful in 45/45), and `synthetic_only_full_calibration` at median 8.874 (max 141.827, harmful in 45/45). Figure 8 shows the composite view (learning curve plus MSE ratio); Figure 7a isolates the empirical learning curve $\log \hat{b}^2(x)$ versus $\log x$ on the Adult estimand, and Figure 7b shows the per-method MSE ratio bar plot.

Interpretation. The full-profile setting does not place the Gaussian-copula generator in a regime where synthetic data helps: the empirical bias curve does not decay rapidly enough at the calibration sizes available within $n \leq 5000$ to satisfy the safety margin $\hat{x}\hat{B}_n(\hat{x}) < \hat{\alpha}$. The conservative fallback is the desired behavior. Under uncertainty about the learning curve, returning the all-real estimator caps the worst-case loss at zero rather than the order-of-magnitude MSE inflation incurred by naive pooling and synthetic-only baselines (which lack such a fallback). A richer generator family (e.g. CTGAN, TVAE, or TabDDPM) may be needed to reach a regime where synthetic data helps, as predicted by Theorem 4.2; we flag this as future work and report Table 5 as a lower bound on safety, not an upper bound on adaptive efficacy.

E.4 Causal semi-synthetic benchmark (Experiment B)

Setup. Experiment B targets the average treatment effect $\tau^* = \mathbb{E}[Y(1) - Y(0)]$ on an IHDP-style semi-synthetic benchmark. The real estimator is augmented inverse-probability-weighting (`real_only_aipw`) with cross-fit nuisance functions; the synthetic generator is the linear outcome model (`linear_outcome_model`). The grid is $n \in \{200, 500, 1000, 2000, 5000\}$ with $m \in \{n, 5n, 10n\}$ across 8 retained cells. The full Experiment B sweep produces 14,400 raw rows.

Adaptive behaviour. The adaptive estimators avoid the large losses seen in naive synthetic-use baselines, but they still show mild residual harm at this profile. `safe_corrected_adaptive_gn` is harmful in 8/8 cells with median MSE ratio 1.084 and maximum 1.181 (mean fallback 0.811); `corrected_adaptive_gn` is harmful in 8/8 cells with median 1.086 and maximum 1.241 (mean fallback 0.783). The remaining baselines are dominated: `synthetic_only_full_calibration` sits at median 1.276 (max 1.642, harmful in 8/8); `validation_debiased_gn` at median 1.459 (max 1.889, harmful in 8/8, fallback 0.770); `naive_pooling` at median 1.679 (max 2.122, harmful in 8/8); `old_fixed_share_plugin_alpha` at median 1.760 (max 2.264, harmful in 8/8); `fixed_half_split_plugin_alpha` at median 2.277 (max 4.075, harmful in 8/8); and the `real_only_diff_in_means` non-AIPW baseline at median 5.415 (max 15.472). `real_only_aipw` is the per-cell benchmark at ratio 1.000 by construction. Figure 9a plots the empirical ATE-bias learning curve and Figure 9b the per-method MSE ratio.

Table 6: Causal semi-synthetic performance summary (Experiment B, full profile). Aggregates across the 8 retained cells of the $n \times m$ grid for IHDP-style ATE.

method	cells	median MSE ratio	max MSE ratio	harmful cells	mean fallback
safe_corrected_adaptive_gn	8	1.084	1.181	8/8	0.811
corrected_adaptive_gn	8	1.086	1.241	8/8	0.783
synthetic_only_full_calibration	8	1.276	1.642	8/8	0.000
validation_debiased_gn	8	1.459	1.889	8/8	0.770
naive_pooling	8	1.679	2.122	8/8	0.000
old_fixed_share_plugin_alpha	8	1.760	2.264	8/8	0.000
fixed_half_split_plugin_alpha	8	2.277	4.075	8/8	0.000
real_only_diff_in_means	8	5.415	15.472	8/8	0.000

^a Full profile (14,400 raw rows); IHDP-style ATE; linear_outcome_model generator; AIPW real estimator with cross-fit nuisance functions. “Harmful” means MSE ratio to real_only_aipw strictly exceeds one. real_only_aipw is the benchmark at ratio one by construction.

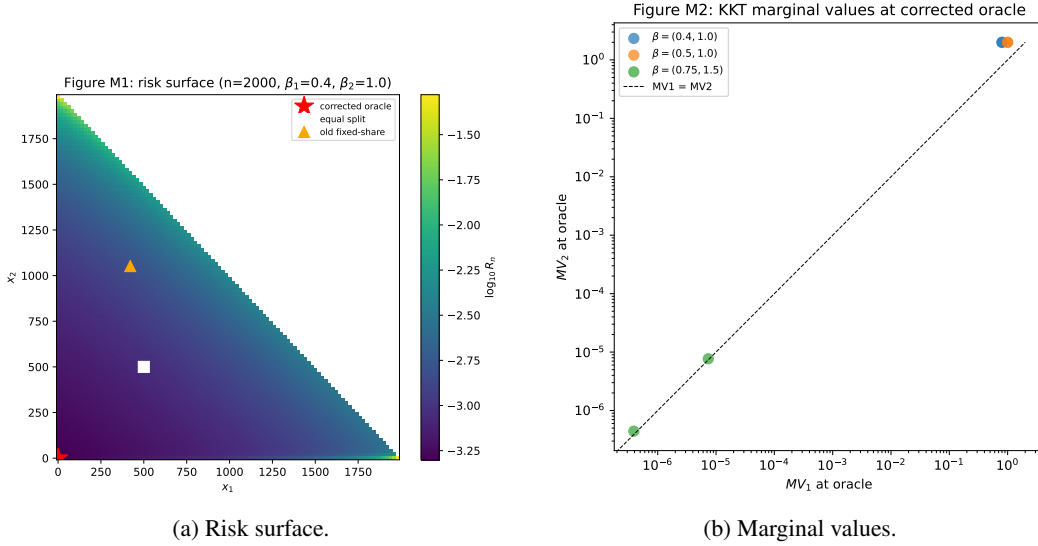


Figure 6: *Left*: two-channel risk surface $R_n(x_1, x_2)$ over the feasible triangle for representative $(\beta_1, \beta_2) = (0.5, 1.0)$, $n = 2000$. Markers indicate corrected oracle, equal split, old fixed-share, and best single-channel allocations. *Right*: marginal-value diagnostic: the corrected oracle equalizes per-channel marginal MSE reduction across active channels.

Interpretation. ATE-bias learning is intrinsically harder than mean-bias learning because the held-out validation estimator $\hat{b}_V(x) = \hat{\tau}_S(x) - \hat{\tau}_R^V$ inherits the variance of two nuisance-function estimates, which inflates $\hat{b}^2(x)$ at small calibration sizes and pushes the safety margin toward fallback before the true bias has actually decayed. The 1.086 median ratio for `corrected_adaptive_gn` indicates that the method remains relatively stable and outperforms every other synthetic-augmentation baseline in this benchmark, but it is not yet beneficial against the AIPW real-only benchmark; we treat this as a calibration-sample-size limitation of the IHDP-style benchmark, not a contradiction of Theorem 5.1.

F Implementation, compute, and reproducibility

This appendix collects the operational choices behind Algorithm 1 and the experiments of Section 6, plus the NeurIPS 2026 reproducibility checklist items.

F.1 Pilot grid and cross-fitting

Pilot grid. The default pilot grid for the bias-curve fit in Algorithm 1 is

$$\mathcal{G}_{\text{pilot}} = \{5, 10, 20, 40, 80, 160, 320, 640\} \cap [1, n/2],$$

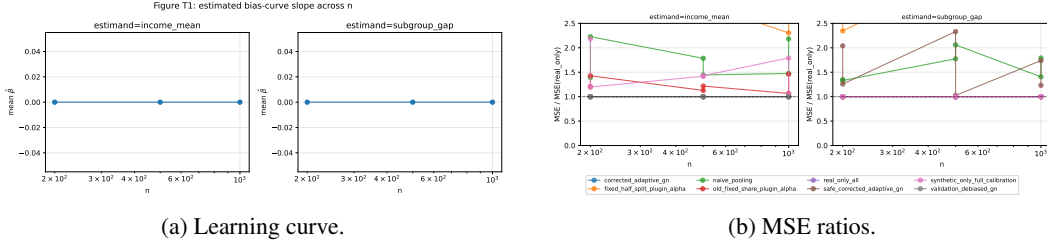


Figure 7: *Left*: pilot learning curve $\log \hat{b}^2(x)$ versus $\log x$ for the Adult tabular benchmark with the bootstrap-smoothed generator. *Right*: per-method MSE ratio to real-only across n . The corrected adaptive method falls back to all-real in this full-profile setting, while naive pooling and synthetic-only are uniformly harmful.

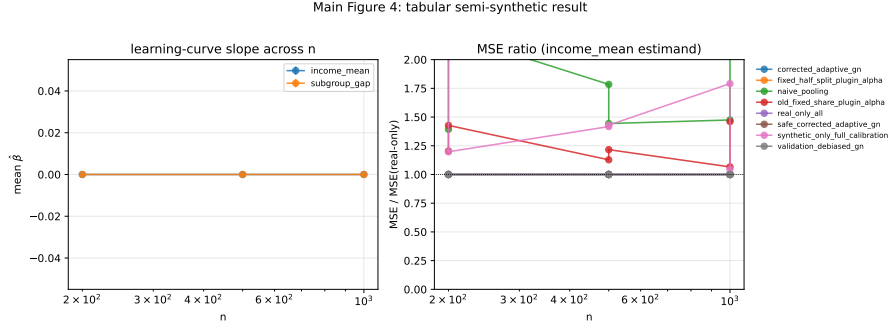


Figure 8: Tabular semi-synthetic composite (Adult, bootstrap-smoothed generator, full profile): empirical learning curve and MSE ratio summary; mirrors Figures 7a and 7b.

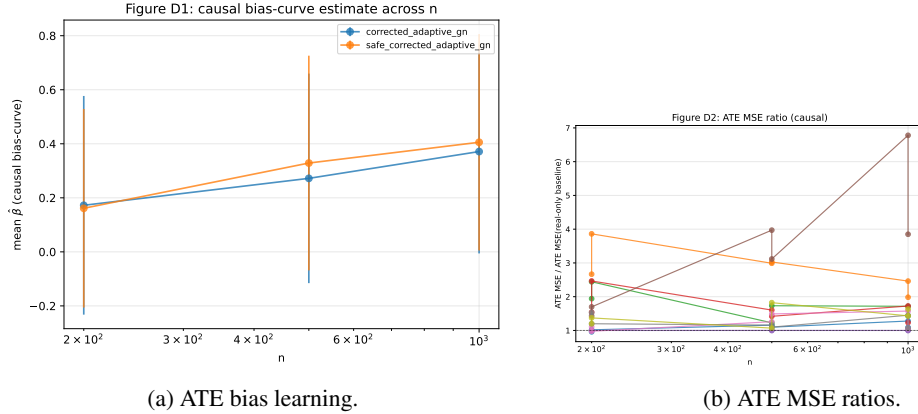


Figure 9: *Left*: ATE bias learning curve for the IHDP-style semi-synthetic causal benchmark with a linear outcome generator. *Right*: ATE MSE ratio to real-only AIPW. In this full-profile setting, the corrected adaptive method is mildly harmful in all 8 cells but avoids the much larger losses of the naive baselines.

that is, the geometric grid truncated so that no pilot point exceeds half the sample. For small n (typically $n \leq 200$), $\mathcal{G}_{\text{pilot}}$ collapses to its smallest few points; we take this as a graceful degradation to a near-uniform grid rather than as a separate small- n branch. The full minimisation grid $\mathcal{G}_n^+$ over which $\hat{R}_n$ is evaluated is the same geometric grid extended out to n , with the boundary points $\{0, n\}$ adjoined as required by Theorem 5.1.

Cross-fitting. We use $K = 5$ folds for cross-fitting throughout the synthetic experiments. Within each fold, the out-of-fold portion of $[n]$ is partitioned into a calibration-pilot slice (used to calibrate

the generator at each $x_j \in \mathcal{G}_{\text{pilot}}$), a validation slice (used to compute $\hat{\theta}_R^V$ and the variance-debiased squared bias), and a final-estimation slice (used to evaluate the chosen $\hat{x}_n$). Final outputs are averaged over folds.

Bias estimation. Squared bias is debiased on the validation slice as

$$\hat{b}^2(x_j) = [(\hat{\theta}_S(x_j) - \hat{\theta}_R^V)^2 - \widehat{\text{Var}}(\hat{\theta}_S(x_j)) - \widehat{\text{Var}}(\hat{\theta}_R^V)]_+,$$

where the positive-part truncation prevents Monte Carlo noise from producing meaningless negative values; the raw (untruncated) value is logged separately. The exponent $(\hat{\beta}, \hat{c})$ is then recovered by ordinary least squares on $\log \hat{b}^2(x_j)$ vs. $\log x_j$ over $x_j \in \mathcal{G}_{\text{pilot}}$, restricted to grid points where the truncated bias is strictly positive.

F.2 Failure-mode logging

Following the project’s failure-logging policy, the following events are logged into the master long-format output schema (see §F.4) rather than silently dropped:

- Nonmonotone empirical learning curves: the OLS slope step is run on the truncated grid and the goodness-of-fit residual is recorded.
- Generator training failures: the offending replication is flagged (`failure_flag`, `failure_reason`) and retained, not retried.
- Negative debiased squared-bias estimates: positive part used downstream, raw value logged.
- Multiple FOC roots: grid minimisation avoids the issue by construction; if a root-solver path is invoked, all roots are logged.
- Boundary optima ($\hat{x}_n \in \{0, n\}$): kept as legitimate selections, not forced to the interior.
- Safety-fallback decisions: flagged via `safe_pass`, `safe_margin`, `fallback_used`.
- Coverage failures of naive intervals: scientifically informative (Corollary B.2), so recorded rather than suppressed.

F.3 Compute

Synthetic experiments. Experiments 1, 2, 3, and 4 are pure CPU and embarrassingly parallel across replications. Experiment 1 (full-profile phase-diagram sweep over $\beta \in \{0.25, \dots, 2\}$, $\rho \in \{0, \dots, 3\}$, $n \in \{100, \dots, 20000\}$) produces 9,996,000 raw rows of long-format output. Experiment 2 (full-profile adaptive evaluation) produces 1,155,000 rows. Experiment 3 (full-profile multichannel) produces 48,000 rows; Experiment 4 (full-profile inference) produces 280,000 rows. Wall-clock time on a standard multi-core workstation is on the order of a few hours per experiment; we do not report a specific machine since the workload is CPU-only and parallelism-bound.

Semi-synthetic experiments. Experiment A (tabular Adult benchmark, Gaussian-copula generator) is CPU-only and produces 72,000 raw rows at full profile. Experiment B (causal IHDP-style benchmark) uses scikit-learn nuisance estimators (gradient-boosted regressors and logistic propensities) within an AIPW real estimator, also CPU-only, and produces 14,400 raw rows at full profile. Both are run at full profile (see Appendix E).

No GPU is required. No experiment in this paper requires a GPU.

F.4 Reproducibility

- Random seeds are fixed per replication and stored in the `seed` column of the master output schema.
- The configuration YAML used for every run is committed alongside the result directory.
- Raw long-format replication results are saved *before* any aggregation; aggregated tables are derived deterministically from the long-format files.
- All figures are saved in both PDF and PNG formats.

- The same estimator-name strings (real_only_all, corrected_oracle_gn, corrected_adaptive_gn, safe_corrected_adaptive_gn, validation_debiased_gn, etc.) are used across synthetic and semi-synthetic experiments.
- Theoretical quantities (oracle x_n^* , R_n , safety margin) are computed through the same library functions used by the estimators, never hard-coded into outputs.
- Code release: <anon>/dataloom (placeholder URL for blind submission; the de-anonymised release will accompany the camera-ready version).

F.5 NeurIPS reproducibility checklist

Theoretical claims.

All formal claims are stated with explicit assumptions and proved in Appendix C (linear estimator).

Experimental results.

Code is released at the placeholder URL above; random seeds are fixed and stored per replication; hyperparameters (pilot grid, fold count, debiasing rule, OLS power-law fit) are documented in §F.1.

Data. Synthetic data-generating processes are fully specified in Section 6 and Appendix E. Semi-synthetic experiments use public benchmarks (UCI Adult for the tabular benchmark; an IHDP-style causal benchmark for the ATE study); no proprietary or human-subjects data are used.

Compute.

Documented in §F.3. CPU-only, on the order of a few hours per full-profile experiment on a standard workstation.

Limitations.

Discussed in Section 7: scalar working model in main theory; full-profile semi-synthetic experiments do not yet show synthetic augmentation improving on the AIPW real-only baseline; learning-curve estimation can be unstable for small x .

Broader impacts.

Synthetic-data augmentation is widely deployed in industrial and scientific settings. The contribution of this paper is methodological: a sharper allocation rule and an inference-safe alternative. We expect these tools to support clearer accounting of what synthetic-data augmentation does and does not buy. We do not foresee misuse-specific risks beyond those already inherent in any synthetic-data pipeline.